

7. Fertilizer Recommendation for Crops

7.1 Cereal Crops

RICE (*Oryza sativa* L.)

Boro rice

(Var: BRRI dhan29, BRRI dhan58, BRRI dhan89, BRRI dhan92, BRRI dhan96, Bangabandhu dhan100, BRRI dhan102, Binadhan-14, Binadhan-18, Binadhan-24, Binadhan-25)

Yield Goal: 7.5 ± 0.75 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)					OF* (t/ha)
	N	P	K	S	Zn	
Optimum	0-58	0-6	0-40	0-6	–	2
Medium	59-116	7-12	41-80	7-12	0.0–1.3	
Low	117-174	13-18	81-120	13-18	1.4–2.6	
Very low	175-232	19-24	121-160	19-24	2.7–3.9	

OF=Organic fertilizers

Boro rice

(Var: BRRI dhan28, BRRI dhan50, BRRI dhan63, BRRI dhan67, BRRI dhan81, BRRI dhan84, BRRI dhan86, BRRI dhan88, Binadhan-8, Binadhan-10)

Yield Goal: 6.0 ± 0.6 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)					OF* (t/ha)
	N	P	K	S	Zn	
Optimum	0-46	2	0-32	0-5	0	2
Medium	47-92	6-10	33-64	6-10	0-1.0	
Low	93-138	11-15	65-96	11-15	1.1-2.0	
Very low	138-184	16-20	97-128	16-20	2.1-3.0	

Method of fertilizer application:

- In general, all of P, K, S and Zn fertilizers should be applied as basal during final land preparation. S and Zn can be topdressed if needed (depending on symptom).
- For light textured soils, half of K should be applied as basal and the rest half should be applied with last N topdressing.
- Nitrogen should be applied in three equal splits, the first one as basal/immediately after seedling establishment, the second one at early tillering stage and the third one at 5–7 days before panicle initiation (PI) stage. It should be applied as broadcast and mixed thoroughly with the soil as soon as possible for better utilization.
- Under direct seeded culture fertilizer should be applied in two equal splits, the first one as basal and the second one at maximum tillering stage; and should be mixed thoroughly with the soil as soon as possible for better utilization.

- e) About 30% urea-N would be saved when USG-N is used. If the dose of N is 150 kg/ha, 110 kg N/ha should be applied as UMG (2.7 g each) and the rest 40 kg N/ha to be top dressed as prilled urea at one week before panicle initiation.

Aus rice

(Var. BRRI dhan48, BRRI dhan82, BRRI dhan85, BRRI dhan98, BRRI dhan106, Binadhan-21)

Yield Goal: 5.0 ± 0.5 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)				
	N	P	K	S	Zn
Optimum	0-24	0-4	0-22	0-3	—
Medium	25-48	5-8	23-44	4-6	0.0-0.7
Low	49-72	9-12	45-66	7-9	0.8-1.4
Very low	73-96	13-16	67-88	10-12	1.5-2.1

Aus rice

(Var. BRRI dhan27, BRRI dhan83, Binadhan-19)

Yield Goal: 4.0 ± 0.4 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)				
	N	P	K	S	Zn
Optimum	0-21	0-3	0-19	0-2.5	—
Medium	22-42	4-6	20-38	2.6-5.0	0.0-0.5
Low	43-63	7-9	39-57	5.1-7.5	0.6-1.0
Very low	64-84	10-12	58-75	7.8-10	1.1-1.5

T. Aman rice

(Var: BRRI dhan72, BRRI dhan73, BRRI dhan87, BRRI dhan93, BRRI dhan94, BRRI dhan95, BRRI dhan103, Binadhan-16, Binadhan-17, Binadhan-22, Binadhan-23)

Yield Goal: 6.0 ± 0.6 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)				
	N	P	K	S	Zn
Optimum	0-30	0-5	0-27	0-4	—
Medium	31-60	6-10	28-54	5-8	0-0.9
Low	61-90	11-15	55-81	9-12	1.0-1.8
Very low	91-120	16-20	82-108	13-16	1.9-2.7

T. Aman rice

(Var: BR22, BR23, BRRI dhan30, BRRI dhan33, BRRI dhan46, BRRI dhan49, BRRI dhan51, BRRI dhan52, BRRI dhan62, BRRI dhan70, BRRI dhan71, BRRI dhan75, BRRI dhan76, BRRI dhan77, BRRI dhan78, BRRI dhan79, Binadhan-7, Binadhan-11, Binadhan-12, Binadhan-15, Binadhan-16, Binadhan-17, Binadhan-20)

Yield Goal: 5.0 ± 0.5 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)				
	N	P	K	S	Zn
Optimum	0-24	0-4	0-22	0-3	–
Medium	25-48	5-8	23-44	4-6	0.0-0.7
Low	49-72	9-12	45-66	7-9	0.8-1.4
Very low	73-96	13-16	67-88	10-12	1.5-2.1

T. Aman rice

(Var: BRRI dhan34, BRRI dhan90, Binadhan-12, Binadhan-13)

Yield Goal: 4.0± 0.4 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)				
	N	P	K	S	Zn
Optimum	0-19	0-3	0-18	0-2.5	–
Medium	20-38	4-6	19-36	2.6-5.0	0.0-0.7
Low	39-57	7-9	37-54	5.1-7.5	0.8-1.4
Very low	58-76	10-12	55-72	7.6-10	1.5-2.1

Method of fertilizer application: Please follow the method as described in page 76.

T. Aman rice

(Variety: BR5, Binadhan-9;
(LIV: Kataribhog, Kalijira, Chinigura)

Yield Goal: 3.0 ± 0.3 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)				
	N	P	K	S	Zn
Optimum	0-18	0-2.5	0-16	0-2.5	–
Medium	19-36	2.6-5.0	17-32	2.6-5.0	0.0-0.6
Low	37-54	5.1-7.5	33-48	5.1-7.5	0.7-1.2
Very low	55-72	7.6-10	49-64	7.6-10	1.3-1.8

B. Aman rice

(Variety: BRRI dhan91)
(Var. LIV under Broadcast Culture)

Yield Goal: 2.5 ± 0.25 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)				
	N	P	K	S	Zn
Optimum	0–12	0-2.5	0–10	0–2	–
Medium	13–24	2.6-5.0	11–20	3–4	0.0–0.5
Low	25–36	5.1-7.5	21–30	5–6	0.6–1.0
Very low	37–48	7.6-10	31–40	7–8	1.1–1.5

Fertilizer recommendation for seedbed

Usually, fertilizer application is not required for rice seedbed. For low fertile soils, manure at the rate of 2 kg/m² can be applied. In case of sulphur deficiency, gypsum should be applied at the rate of 10 g/m² as topdressing. Yellow coloured seedlings due to cold injury in Boro season cannot be recovered by N or S topdressing, rather the seedbed should be covered with transparent polythene.

WHEAT (*Triticum aestivum*)

[Var: BARI Gom 25, BARI Gom 28, BARI Gom 30, BARI Gom 32, BARI Gom 33, BWMRI Gom 1, BWMRI Gom 2, BWMRI Gom 3, BWMRI Gom 4, BWMRI Gom 5.]

Yield Goal: 4.5 ± 0.45 t/ha (Irrigated culture)

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)							OF* (t/ha)
	N	P	K	S	Mg	Zn	B	
Optimum	0–40	0-10	0–30	0–5	–	–	–	2
Medium	41–80	11-20	31–60	6–10	0–4	0.0–1.3	0.0–1.0	
Low	81–120	21-30	61–90	11–15	4–8	1.4–2.6	1.1–2.0	
Very low	121–160	31-40	91–120	16–20	7–12	2.7–3.9	2.1–3.0	

Method of application (Irrigated culture):

- Two-thirds of the N and all of P, K, S, Mg, Zn, B and organic manure (if used) should be applied as basal during final land preparation.
- Remaining one-third N should be applied after 17–21 days of sowing (DAS) after first irrigation.

MAIZE (*Zea mays*)

(Var: BARI Hybrid Bhutta-9, BARI Hybrid Bhutta-13, BARI Hybrid Bhutta-14, BARI Hybrid Bhutta-15, BARI Hybrid Bhutta-16, BARI Hybrid Bhutta-17, BWMRI Hybrid Maize-1, BWMRI Hybrid Maize-2)

Yield Goal: 11.0 ±1.1 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)							OF* (t/ha)
	N	P	K	S	Mg	Zn	B	
Optimum	0–80	0–18	0–45	0–12	0–5	–	–	3
Medium	81–160	19–36	46–90	13–24	6–10	0.0–2.0	0.0–1.0	
Low	161–240	37–54	91–135	25–36	11–15	2.1–4.0	1.1–2.0	
Very low	241–320	55–72	136–180	37–48	16–20	4.1–6.0	2.1–3.0	

Method of application:

Rabi season:

- One-third of nitrogen and all of P, K, S, Zn, B, and organic manure (if any) should be applied during sowing in 7–10 cm deep furrows (7–10 cm apart from the rows) and covered with the soil.
- Half of the remaining N should be applied at 50–55 DAS and the rest half should be applied at tasselling stage (80–85 DAS) as side dressing immediately after irrigation.

Kharif season:

- For Kharif season the dose may be reduced by 30%. All of P, K, S, Mg, Zn and B should be applied during sowing in 7–10 cm deep furrows (7–10 cm apart from the rows) and covered with the soil.
- The remaining N should be applied in two equal splits as side dressing in rows at 8–10 leaf stage (20–25 DAS) and at tasseling stage (45–50 DAS) and mixed thoroughly with the soil.

BARLEY (*Hordeum vulgare*)

(Var. BARI Barley-1, BARI Barley-2, BARI Barley-3, BARI Barley-4, BARI Barley-5, BARI Barley-6, BARI Barley-7, BARI Barley-8, BARI Barley-9)

Yield Goal: 3.0 ± 0.3 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)				
	N	P	K	S	Zn
Optimum	0–20	0–7	0–15	0–3	–
Medium	21–40	8–14	16–30	4–6	0.0–1.0
Low	41–60	15–21	31–45	7–9	1.1–2.0
Very low	61–80	22–28	46–60	10–12	2.1–3.0

Method of application:

- Half of N and all of P, K, S and Zn should be applied as basal during final land preparation.
- Remaining N should be applied as topdress followed by light irrigation at 55-60 DAS for barley and foxtail millet and 30-35 DAS for proso millet.
- Under rainfed culture all fertilizers should be applied as broadcast during final land preparation.

PROSO MILLET (*Panicum miliaceum*)

(Var. BARI Cheena-1)

Yield Goal: 2.5 ± 0.25 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)				
	N	P	K	S	Zn
Optimum	0–20	0–7	0–15	0–3	–
Medium	21–40	8–14	16–30	4–6	0.0–1.0
Low	41–60	15–21	31–45	7–9	1.1–2.0
Very low	61–80	22–28	46–60	10–12	2.1–3.0

Method of application: as mentioned in barley.

FOXTAIL MILLET (*Setaria italica*)

(Var. Titas, BARI Kaon-2 and BARI Kaon-3, BARI Kaon-4)

Yield Goal: 3.0 ± 0.3 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)				
	N	P	K	S	Zn
Optimum	0–20	0–8	0–14	0–3	–
Medium	21–40	9–16	15–28	4–6	0.0–1.0
Low	41–60	17–24	29–42	7–9	1.1–2.0
Very low	61–80	25–32	43–56	10–12	2.1–3.0

Method of application: Same as barley.

7.2 Fibre Crops

JUTE (*Corchorus capsularis*)

(Var. CVL 1, CVE 3, CC 45, BJRI Deshi Pat 5, BJRI Deshi Pat 6, BJRI Deshi Pat 7, BJRI Deshi Pat 8, BJRI Deshi Pat 9, BJRI Deshi Pat 10, BJRI Deshi Pat shak 1, Binadeshpat-2, Atompat-38)

Yield Goal : 3.5 ± 0.35 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)				
	N	P	K	S	Zn
Optimum	0-30	0-4	0-22	0-6	–
Medium	31-60	5-8	23-44	7-12	0.0–1.5
Low	61-90	9-12	45-66	13-18	1.6–3.0
Very low	91-120	13-16	67-88	19-24	3.1–4.5

Method of application:

- Half of N and all of P, K, S, Zn and organic fertilizer (if any) should be applied as basal during final land preparation.
- Remaining N should be top dressed at 40–45 DAS under moist soil condition.
- Application of 3 t/ha organic fertilizer is recommended; if applied, the dose of N, P, K and S should be reduced based on the rate of application of organic manure (as per Appendix-6).

JUTE (*Corchorus olitorius*)

(Var. O 4, O 9897, BJRI Tossa Pat 4, BJRI Tossa Pat 5, BJRI Tossa Pat 6, BJRI Tossa Pat 7, BJRI Tossa Pat 8, BJRI Tossa Pat 9)

Yield Goal: 4.4 ± 0.44 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)				
	N	P	K	S	Zn
Optimum	0-35	0-6	0-25	0-6	–
Medium	36-70	7-12	26-50	7-12	0.0-2.0
Low	71-105	13-18	51-75	13-18	2.1-4.0
Very low	106-140	19-24	76-100	19-24	4.1-6.0

Method of application: Same as jute (*Corchorus capsularis*)

KENAF (*Hibiscus cannabinus*)

(Var. HC 2, BJRI Kenaf 2, BJRI Kenaf 3, BJRI Kenaf 4)

Yield Goal: 3.5 ± 0.35 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)				
	N	P	K	S	Zn
Optimum	0-30	0-5	0-22	0-6	–
Medium	31-60	6-10	23-44	7-12	0.0-1.5
Low	61-90	11-15	45-66	13-18	1.6-3.0
Very low	91-120	16-20	67-88	19-24	3.1-4.5

Method of application: Same as jute (*Corchorus capsularis*)

MESTA (*Hibiscus sabdariffa*)

(Var. HS 24, BJRI Mesta 2, BJRI Mesta 3, BJRI Mesta 4)

Yield Goal : 3.5± 0.35 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)				
	N	P	K	S	Zn
Optimum	0-27	0-4	0-20	0-6	—
Medium	28-54	5-8	21-40	7-12	0.0-1.5
Low	55-81	9-12	41-60	13-18	1.6-3.0
Very low	82-108	13-16	61-80	19-24	3.1-4.5

Method of application: Same as jute (*Corchorus capsularis*)

LATE JUTE SEED PRODUCTION

JUTE (*Corchorus capsularis*)

(Var. CVL 1, BJRI Deshi Pat 5, BJRI Deshi Pat 6, BJRI Deshi Pat 7, BJRI Deshi Pat 8)

Yield Goal (Seed): 900-1000 kg/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)						OF* (t/ha)
	N	P	K	S	Zn	B	
Optimum	0-29	0-4	0-20	0-6	—	0	3
Medium	30-58	5-8	21-40	7-12	0.0-1.5	0-1.0	
Low	59-87	9-12	41-60	13-18	1.6-3.0	1.1-2	
Very low	88-116	13-16	61-80	19-24	3.1-4.5	2.1-3	

*OF: Organic Fertilizer

Method of application:

- All of P, K, S, Zn and organic fertilizer (if any) should be applied, as basal during final land preparation.
- N fertilizer should be applied as topdressing in three equal splits at final land preparation, 20-25 DAS and 40-45 DAS.
- Application of 3 t/ha OF is recommended; if applied, the dose of N, P, K and S should be reduced based on the rate of application of organic manure (as per Appendix-6).

JUTE (*Corchorus olitorius*)

(Var. O 9897, BJRI Tossa Pat 4, BJRI Tossa Pat 8, BJRI Tossa Pat 9)

Yield Goal (seed): 900-1000 kg/ha.

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)						OF (t/ha)
	N	P	K	S	Zn	B	
Optimum	0-30	0-5	0-20	0-5	–	–	3
Medium	31-60	6-10	21-40	6-10	0.0-1.5	0.0-1.0	
Low	61-90	11-15	41-60	11-15	1.6-3.0	1.1-2.0	
Very low	91-120	16-20	61-80	16-20	3.1-4.5	2.1-3.0	

Method of application: Same as jute (*Corchorus capsularis*) for seed production

KENAF (*Hibiscus cannabinus*)

(Var. HC 2, BJRI Kenaf 2, BJRI Kenaf 3, BJRI Kenaf 4)

and

MESTA (*Hibiscus sabdariffa*)

(Var. HS 24, BJRI Mesta 2, BJRI Mesta 3, BJRI Mesta 4)

Yield Goal (Seed): 800-1000kg/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)						OF (t/ha)
	N	P	K	S	Zn	B	
Optimum	0-30	0-5	0-20	0-5	–	–	3
Medium	31-60	6-10	21-40	6-10	0.0-1.5	0.0-1.0	
Low	61-90	11-15	41-60	11-15	1.6-3.0	1.1-2.0	
Very low	91-120	16-20	61-80	16-20	3.1-4.5	2.1-3.0	

Method of application: Same as jute (*Corchorus capsularis*) for seed production.

COTTON (*Gossypium harbaceum*)

(Var. CB-9, CB-10)

Yield Goal: 2.25± 0.23 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)						OF (t/ha)
	N	P	K	S	Zn	B	
Optimum	0–30	0–12	0–25	0–6	-	-	3
Medium	31–60	13–24	26–50	7–12	0.0–1.5	0–0.5	
Low	61–90	25–36	51–75	13–18	1.6–3.0	0.6–1.0	
Very low	91–120	37–48	76–100	19–24	3.1–4.5	1.1–1.5	

Method of application:

- a) **Basal application:** All of FYM should be applied as basal during final land preparation. Of the recommended inorganic fertilizer nutrients, 10% N, 75% P, 15% K, 30% S and 10% B should be applied during sowing in 7-10 cm deep furrows 5-8 cm apart from the seed rows and covered with the soil.
- b) **Side dressing:** Fertilizers should be applied in four installments. Of the recommended inorganic fertilizer nutrients -
 - 20% N, 15% K and 30% Zn should be applied as side dressing during 20-25 days after sowing (DAS).
 - 25% N, 25% P, 30% K, 40% S, 40% Zn and 50% B should be applied as side dressing during 40-45 DAS.
 - 30% N, 30% K, 30% S, 30% Zn and 40% B should be applied as side dressing during 60-65 DAS.
 - 15% N and 10% K should be applied as side dressing during 75-80 DAS.
 - After application the fertilizers should be mixed thoroughly with the soil.

COTTON (*Gossypium harbaceum*)

(Var. CB-12, CB-13, CB-14)

Yield Goal: 2.70± 0.27 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)							OF (t/ha)
	N	P	K	S	Zn	B	Mg	
Optimum	0–40	0–15	0–45	0–8	-	-	0–5	3
Medium	41–80	16–30	46–90	9–16	0–2	0.0–1.0	6–10	
Low	81–120	31–45	91–135	17–24	3–4	1.1–2.0	11–15	
Very low	121–160	46–60	136–180	25–32	5–6	2.1–3.0	16–20	

Method of application: Same as for the varieties CB-9 and CB-10.

COTTON (*Gossypium arboreum*)

(Var. Rupali-1, DM-2, DM-3)

Yield Goal: 3.6 ± 0.36 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)							FYM (t/ha)
	N	P	K	S	Zn	B	Mg	
Optimum	0–45	0–15	0–50	0–8	-	-	0–5	3
Medium	46–90	16–30	51–100	9–16	0–2	0.0–1.0	6–10	
Low	91–135	31–45	101–150	17–24	3–4	1.1–2.0	11–15	
Very low	136–180	46–60	151–200	25–32	5–6	2.1–3.0	16–20	

*FYM: Farm Yard Manure

Method of application: Same as varieties CB-9 and CB-10.

7.3 Pulse Crops

LENTIL (*Lens culinaris*)

(Var. BARI Masur-3, BARI Masur-6, BARI Masur-7, BARI Masur-8, BARI Masur-9, Binamasur-2, Binamasur-3, Binamasur-4, Binamasur-5 and Binamasur-6, Binamasur-7, Binamasur-8, Binamasur-9, Binamasur-10)

Yield Goal: 1.8 ± 0.18 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)						
	N	P	K	S	Zn	B	Mo
Optimum	0–7	0–5	0–8	0–4	-	-	-
Medium	8–14	6–10	9–16	5–8	0–1.0	0–0.4	0–0.4
Low	15–21	11–15	17–24	9–12	1.1–2.0	0.5–0.8	0.5–0.8
Very low	22–28	16–20	25–32	13–16	2.1–3.0	0.9–1.2	0.9–1.2

Method of application:

- a) All fertilizers should be applied as basal during final land preparation.
- b) *Rhizobium* inoculum (at the rate of 50 g/kg seed or 1.5 kg/ha) should be used if available and in that case N fertilizer should not be used.

CHICKPEA (*Cicer arietinum*)

(Var. BARI Chola-5, BARI Chola-6, BARI Chola-7, BARI Chola-9, BARI Chola-10, BARI Chola-11, Binasola-6, Binasola-7, Binasola-8, Binasola-9, Binasola-10)

Yield Goal: 2.0 ± 0.2 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)						
	N	P	K	S	Zn	B	Mo
Optimum	0–9	0–6	0–8	0–4	–	–	–
Medium	10–18	7–12	9–16	5–8	0.0–1.0	0.0–0.6	0–0.4
Low	19–27	13–18	17–24	9–12	1.1–2.0	0.7–1.2	0.5–0.8
Very low	28–36	19–24	25–32	13–16	2.1–3.0	1.3–1.8	0.9–1.2

Method of application:

- All fertilizers should be applied as basal during final land preparation.
- Rhizobium* inoculum (at the rate of 50 g/kg seed or 1.5 kg/ha) should be used if available and in that case N fertilizer should not be used.

MUNGBEAN (*Vigna radiata*)

(Var. BARI Mung-4, BARI Mung-5, BARI Mung-6, BARI Mung-7, BARI Mung-8, Binamoog-5, Binamoog-6, Binamoog-7, Binamoog-8, Binamoog-9, BAU Mung-1, BU Mung-1, BU Mung-2 and BU Mung-4)

Yield Goal: 2.0 ± 0.2 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)						
	N	P	K	S	Zn	B	Mo
Optimum	0–6	0–5	0–8	0–4	–	–	–
Medium	7–12	6–10	9–16	5–8	0.0–1.0	0–0.4	0–0.4
Low	13–18	11–15	17–24	9–12	1.1–2.0	0.5–0.8	0.5–0.8
Very low	19–24	16–20	25–32	13–16	2.1–3.0	0.9–1.2	0.9–1.2

Method of application:

- All fertilizers should be applied as basal during final land preparation.
- Rhizobium* inoculum (at the rate of 50 g/kg seed or 1.5 kg/ha) should be used if available and in that case N fertilizer should not be used.

BLACKGRAM (*Vigna mungo*)

(Var. BARI Mash-1, BARI Mash-2, BARI Mash-3, BARI Mash-4, BINAmash-1)

Yield Goal: 1.5 ± 0.15 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)			
	N	P	K	S
Optimum	0–6	0–5	0–6	0–3
Medium	7–12	6–10	7–12	4–6
Low	13–18	11–15	13–18	7–9
Very low	19–24	16–20	19–24	10–12

Method of application:

- All fertilizers should be applied as basal during final land preparation.
- Rhizobium* inoculum (at the rate of 50 g/kg seed or 1.5 kg/ha) should be used if available and in that case N fertilizer should not be used.

GRASSPEA (*Lathyrus sativus*)

(Var. BARI Khesari-1, BARI Khesari-2, BARI Khesari-3, BARI Khesari-4, BARI Khesari-5, BARI Khesari-6, Binakhesari-1)

Yield Goal: 1.5 ± 0.15 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)*			
	N	P	K	S
Optimum	0–5	0–5	0–6	0–3
Medium	6–10	6–10	7–12	4–6
Low	11–15	11–15	13–18	7–9
Very low	16–20	16–20	19–24	10–12

* Same recommendation for the crop under tillage and no tillage conditions.

Method of application:

For crop with tillage:

- All of N, P, K and S should be applied as basal during final land preparation.
- Rhizobium* inoculum (at the rate of 40 g/kg seed or 2 kg/ha) must be used if available and in that case N fertilizer should not be used.

For relay crop with no tillage:

- All of P, K and S should be applied as basal in the standing aman rice crop at 2-3 days before sowing and nitrogen should be topdressed of aman rice of 10-15 days after sowing of grasspea seed.

COWPEA (*Vigna unguiculata*)

(Var. BARI Falon-1, BARI Falon-2, others)

Yield Goal: 1.4 ± 0.14 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)			
	N	P	K	S
Optimum	0–5	0–5	0–6	0.0–3.0
Medium	6–10	6–10	7–12	3.1–6.0
Low	11–15	11–15	13–18	6.1–9.0
Very low	16–20	16–20	19–24	9.1–12.0

Method of application:

- a) All fertilizers should be applied as basal during final land preparation.
- b) Rhizobium inoculum (at the rate of 40 g/kg seed or 1.6 kg/ha) should be used if available and in that case N fertilizer should not be used.

7.4 Oil Seed Crops

MUSTARD

(*Brassica juncea*, *Brassica napus*)

(Var. BARI Sharisa-11, BARI Sharisa-13, BARI Sharisa-16, BARI Sharisa-17, BARI Sharisa-18 (Canola), BARI Sharisa-19, Binasharisa-3, Binasharisa-4 and Binasharisa-5, Binasharisa-7, Binasharisa-9 and Binasharisa-10)

Yield Goal: 2.0 ± 0.2 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)							OF (t/ha)
	N	P	K	S	Mg	Zn	B	
Optimum	0-30	0-8	0-25	0-8	—	—	—	3
Medium	31-60	9-16	26-50	9-16	0-3	0-1.3	0.0-1.0	
Low	61-90	17-24	51-75	17-24	4-6	1.4-2.6	0.6-2.0	
Very low	91-120	25-32	76-100	25-32	7-9	2.7-3.9	1.1-3.0	

Method of application:

- a) Half of N and all of P, K, S, Mg, Zn, B and OF should be applied as basal during final land preparation. Remaining half of N should be applied as top dress at the time of flower initiation (25 days after sowing).
- b) Under rainfed condition, all fertilizers should be applied as basal during final land preparation.
- c) Organic fertilizer, if applied, the dose of N, P, K and S should be reduced based on the rate of application of organic manure (as per Appendix-6).

Mustard (*Brassica campestris*)

(Var. Tori-7, BARI Sharisa-14, BARI Sharisa-15, BARI Sharisa-20, Safal, Agrani, Binasharisa-6, Binasharisa-7, Binasharisa-8)

Yield Goal: 1.8 ± 0.18 t/ha t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)							OF (t/ha)
	N	P	K	S	Mg	Zn	B	
Optimum	0–30	0–8	0–20	0–5	–	–	–	3
Medium	31–60	9–16	21–40	6–10	0.0–2.5	0.0–1.0	0.0–0.5	
Low	61–90	17–24	41–60	11–15	2.6–5.0	1.1–2.0	0.6–1.0	
Very low	91–120	25–32	61–80	16–20	5.1–7.5	2.1–3.0	1.1–1.5	

Method of application: Same as *Brassica juncea*

SESAME (*Sesamum indicum*)

(Var: BARI Til-4, BARI Til-5 BARI Til-6, Binatil-1, Binatil-2, Binatil-3, Binatil-4)

Yield Goal: 1.4± 0.14 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)					
	N	P	K	S	Zn	B
Optimum	0–25	0–8	0–27	0–3	-	-
Medium	26–50	9–16	28–54	4–6	0–1.0	0–0.8
Low	51–75	17–24	55–81	7–9	1.1–2.0	0.9–1.6
Very low	76–100	25–32	82–108	10–24	2.1–3.0	1.7–2.4

Method of application:

- Half of N and all of P, K, S, Zn and B should be applied as basal during final land preparation. Remaining half of N should be applied as top dress at 25–30 DAS.
- Under rainfed condition, all fertilizers should be applied as basal during final land preparation.

GROUNDNUT (*Arachis hypogaea*)

(Var. Jhingabadam, Tridanabadam, BARI Chinabadam-7, BARI Chinabadam-8, BARI Chinabadam-9, BARI Chinabadam-10, BARI Chinabadam-11, BARI Chinabadam-12, Binachinabadam-4, Binachinabadam-5, Binachinabadam-6, Binachinabadam-7, Binachinabadam-8, Binachinabadam-9)

Yield Goal: 2.6 ± 0.26 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)						
	N	P	K	S	Zn	B	Mo
Optimum	0–12	0–10	0–20	0–10	–	–	–
Medium	13–24	11–20	21–40	11–20	0–1.3	0.0–1.0	0–0.4
Low	25–36	21–30	41–60	21–30	1.4–2.6	0.6–2.0	0.5–0.8
Very low	37–48	31–40	61–80	31–40	2.7–3.9	1.1–3.0	0.9–1.2

Method of application:

- Half of N and all of P, K, S, Zn, B and Mo should be applied as basal during final land preparation. Remaining N should be applied as top dress at flowering stage and mixed thoroughly with the soil as soon as possible for better utilization.
- Rhizobium* inoculum (at the rate of 30 g/kg seed) must be used if available. In that case N fertilizer should not be used.
- Under rainfed condition all fertilizers should be applied as basal during final land preparation.

SOYBEAN (*Glycine max*)

(Var. BARI Soybean-6, BARI Soybean-7, Binasoybean-3, Binasoybean-4, Binasoybean-5)

Yield Goal: 2.0 ± 0.2 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)							OF (t/ha)
	N	P	K	S	Zn	B	Mo	
Optimum	0–9	0–12	0–20	0–7	–	–	–	2
Medium	10–18	13–24	21–40	8–14	0–1.0	0.0–0.5	0–0.2	
Low	19–27	25–36	41–60	15–21	1.0–2.0	0.6–1.0	0.3–0.4	
Very low	28–36	37–48	61–80	22–24	2.0–3.0	1.1–1.5	0.5–0.6	

Method of application:

- All fertilizers should be applied as basal during final land preparation.
- Inoculum (at the rate of 20 g/kg seed) should be used if available and in that case N fertilizer should not be used.
- Application of OF at the rate of 2 t/ha is recommended; if applied the dose of N, P, K and S should be reduced based on the rate of application of organic fertilizer (as per Appendix-6).

SUNFLOWER (*Helianthus annus*)

(Var. BARI Surjamukhi-2, BARI Surjamukhi-3)

Yield Goal: 2.5± 0.25 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)							OF (t/ha)
	N	P	K	S	Mg	Zn	B	
Optimum	0–35	0–12	0–30	0–10	—	—	—	2
Medium	36–70	13–24	31–60	11–20	0.0–3.0	0.0–1.0	0.0–0.8	
Low	71–105	25–36	61–90	21–30	4.0–6.0	1.1–2.0	0.9–1.6	
Very low	106–140	37–48	91–120	31–40	7.0–9.0	2.1–3.0	1.7–2.4	

Method of application:

- Half of N and all P, K, S, Mg, Zn, B and OF should be applied as basal during final land preparation. Remaining half of N should be applied as top dress in two equal splits at 20–25 DAS and 40–45 DAS (before flower initiation stage) and mixed thoroughly with the soil as soon as possible for better utilization.
- Application of OF at the rate of 2 t/ha is recommended. In that case the dose of N, P, K and S should be reduced based on the rate of application of organic manure (as per Appendix-6).

SAFFLOWER (*Carthenus tintorious*)

(Var. BARI Saff-1)

Yield Goal: 1.5 ± 0.15 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)					
	N	P	K	S	Zn	B
Optimum	0–25	0–10	0–20	0–7	—	—
Medium	26–50	11–20	21–40	8–14	0.0–1.3	0.0–1.0
Low	51–75	21–30	41–60	15–21	1.4–2.6	1.1–2.0
Very low	76–100	31–40	61–80	22–28	2.7–3.9	2.1–3.0

Method of application:

- Half of N and all of P, K, S, Zn and B should be applied as basal during final land preparation. Remaining N should be applied at the time of flower initiation as top dress.
- Under rainfed condition all fertilizers should be applied as basal during final land preparation.

LINSEED (*Linum usitatissimum*)

(Var. BARI Tishi-1 (Neela), BARI Tishi-2)

Yield Goal: 1.0 ± 0.1 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)			
	N	P	K	S
Optimum	0–15	0–5	0–8	0–3
Medium	16–30	6–10	9–16	4–6
Low	31–45	11–15	17–24	7–9
Very low	46–60	16–20	25–32	10–12

Method of application:

- Half of N and all of P, K, S, Zn and B should be applied as basal during final land preparation. Remaining N should be applied at the time of flower initiation as top dress.
- Under rainfed condition all fertilizers should be applied as basal during final land preparation.

NIGER (*Guizotia abyssinica*)

(Var: BARI Guji-1)

Yield Goal: 1.5 ± 0.15 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)					
	N	P	K	S	Zn	B
Optimum	0–25	0–10	0–20	0–7	–	–
Medium	26–50	11–20	21–40	8–14	0.0–1.3	0.0–1.0
Low	51–75	21–30	41–60	15–21	1.4–2.6	1.1–2.0
Very low	76–100	31–40	61–80	22–28	2.7–3.9	2.1–3.0

Method of application:

- Half of N and all of P, K, S, Zn and B should be applied as basal during final land preparation. Remaining N should be applied at the time of flower initiation as top dress.
- Under rainfed condition all fertilizers should be applied as basal during final land preparation.

7.5 Root and Tuber Crops

POTATO (*Solanum tuberosum*)

(**Var.** BARI Alu-7 (Diamant), BARI Alu-8 (Cardinal), BARI BARI Alu-13 (Granola), BARI Alu-25, BARI Alu-28, BARI Alu-29, BARI Alu-36, BARI Alu-37, BARI Alu-40, BARI Alu-41, BARI Alu-44, BARI Alu-48, BARI Alu-51, BARI Alu-52, BARI Alu-54 (Musica), BARI Alu-56, BARI Alu-62, BARI Alu-72, BARI Alu-73, BARI Alu-77, BARI Alu-78, BARI Alu-79, BARI Alu-85, BARI Alu-86, BARI Alu-90, BARI Alu-91)

Yield Goal: 30.0± 3.0 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)							OF (t/ha)
	N	P	K	S	Mg	Zn	B	
Optimum	0–45	0–10	0–45	0–5	–	–	–	3
Medium	46–90	11–20	46–90	6–10	0–5	0–2.0	0–0.8	
Low	91–135	21–30	91–135	11–15	6–10	3.0–4.0	0.81–1.6	
Very low	136–180	31–40	136–180	16–20	11–15	5.0–6.0	1.61–2.4	

Method of application:

- All of OF, P, K, S, Mg, Zn and B and half of N and K should be applied as basal during final land preparation.
- Remaining half for both N and K should be applied as side dressing at 30–35 days after planting during earthing up operation.

SWEET POTATO (*Ipomoea batatas*)

(**Var.** BARI Misti Alu-8, BARI Misti Alu-10, BARI Misti Alu12, BARI Misti Alu-14, BARI Misti Alu-15 BARI Misti Alu-16, BARI Misti Alu-17)

Yield Goal: 40.0 ± 4.0 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)							OF (t/ha)
	N	P	K	S	Mg	Zn	B	
Optimum	0–40	0–12	0–40	0–5	–	–	–	3
Medium	41–80	13–24	41–80	6–10	0–5	0.0–1.5	0.0–0.8	
Low	81–120	25–36	81–120	11–15	6–10	1.6–3.0	0.9–1.6	
Very low	121–160	37–48	121–160	16–20	11–15	3.1–4.5	1.7–2.4	

Method of application:

- All P, S, Mg, Zn and B and half of N and K should be applied as basal during final land preparation. Entire amount of organic fertilizer (OF) should also be applied during final land preparation.
- Remaining half for both N and K should be applied as side dressing at 30–35 days after planting during earthing up operation.
- Under rainfed condition all fertilizers should be applied during final land preparation.

AROIDS**MUKHI KACHU (*Colocasia esculenta*)**

(Var. Bilasi, BARI Mukhikachu-2, BARI Mukhikachu-3)

Yield Goal: 30.0± 3.0 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)						OF (t/ha)
	N	P	K	S	Zn	B	
Optimum	0-30	0-11	0-30	0-5	—	—	3
Medium	31-60	12-22	31-60	6-10	0.0-1.0	0.0-0.7	
Low	61-90	23-33	61-90	11-15	1.1-2.0	0.8-1.4	
Very low	91-120	34-44	91-120	16-20	2.1-3.0	1.5-2.1	

Method of application: Same as Panikachu**PANI KACHU (*Colocasia esculenta*)**

(Var. Var. BARI Panikachu-1 (Latiraj), BARI Panikachu-2, BARI Panikachu-3, BARI Panikachu-4, BARI Panikachu-5, BARI Panikachu-6 and BARI Panikachu-7, BARI Panikachu-8, BARI Panikachu-9)

Yield Goal: 30.0± 3.0 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)						OF (t/ha)
	N	P	K	S	Zn	B	
Optimum	0-40	0-10	0-45	0-5	—	—	3
Medium	41-80	11-20	61-90	6-10	0.0-1.0	0.0-0.7	
Low	81-120	21-30	121-135	11-15	1.1-2.0	0.8-1.4	
Very low	121-160	31-40	181-180	16-20	2.1-3.0	1.5-2.1	

Method of application:

- Full dose of OF, P, S and half of K potassium should be applied as basal during final land preparation.
- The remaining K should be applied at 60 days after planting.
- The nitrogen should be applied in six splits, starting at 30 days after planting and then at 15-20 days intervals.

7.6 Vegetable Crops

CAULIFLOWER (*Brassica oleracea* var. *botrytis*)

(Var. BARI Fulkopi-1, BARI Fulkopi-2, BARI Fulkopi-3)

Yield Goal: 60.0 ± 6.0 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)							OF (t/ha)
	N	P	K	S	Mg	Zn	B	
Optimum	0-40	0-15	0-30	0-6	0-2.5	—	-	5
Medium	41-80	16-30	31-60	7-12	2.6-5.0	0-1.5	0-0.8	
Low	81-120	31-45	61-90	13-18	5.1-7.5	1.6-3.0	0.9-1.6	
Very low	121-160	46-60	91-120	19-24	7.6-10	3.1-4.5	1.7-2.4	

Method of application:

- Half of OF, K and all of P, S, Zn and B should be applied as basal. Remaining half of OF should be applied in pit before planting of seedlings.
- Full N and half of K should be applied in three equal splits at 10-15, 30 and 50 days after transplanting as ring method under moist soil condition and mixed thoroughly with the soil.
- Instead of urea, Urea Supper Granules (USG) can be applied 8–10 cm deep into the soil and 8–10 cm apart from the plant and as ring method.

BROCCOLI (*Brassica oleracea* var. *italica*)

(Var. BARI Brokoli-1)

Yield Goal: 40.0 ± 4.0 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)							OF (t/ha)
	N	P	K	S	Mg	Zn	B	
Optimum	0-30	0-12	0-25	0-5	0-2.5	—	—	5
Medium	31-60	13-24	26-50	6-10	2.6-5.0	0-1.5	0-0.8	
Low	61-80	25-36	51-75	11-15	5.1-7.5	1.6-3.0	0.9-1.6	
Very low	91-120	37-48	76-100	16-20	7.8-10	3.1-4.5	1.7-2.4	

Method of application: Same as cauliflower

CABBAGE (*Brassica oleracea var. capitata*)

(Var. K-K Cross, Atlas-70, BARI Badhacopy-2, Hybrids)

Yield Goal: 90.0 ± 9.0 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)							OF (t/ha)
	N	P	K	S	Mg	Zn	B	
Optimum	0-50	0-15	0-20	0-5	0-2.5	–	-	5
Medium	61-100	16-30	21-40	6-10	2.6-5.0	0-1.5	0-0.8	
Low	101-150	31-45	41-60	11-15	5.1-7.5	1.6-3.0	0.9-1.6	
Very low	151-200	46-52	61-100	16-32	7.8-10	3.1-4.5	1.7-2.4	

Method of application: Same as cauliflower

CHINESE CABBAGE (*Brassica chinensis*)

(Var. BARI Chinakopi-1)

Yield Goal: 50.0 ± 5.0 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)							OF (t/ha)
	N	P	K	S	Mg	Zn	B	
Optimum	0-38	0-11	0-15	0-4	0-1.9	-	-	5
Medium	39-76	12-22	16-30	5-8	2.0-3.8	0-1.4	0-0.8	
Low	77-114	23-33	31-45	9-12	3.9-5.7	1.5-2.8	0.9-1.6	
Very low	115-152	34-44	46-60	13-16	5.8-7.6	2.9-4.2	1.7-2.4	

Method of application:

- Half of OF, K and all of P, S, Zn and B should be applied as basal. Remaining half of OF should be applied in pit before planting of seedlings.
- Full N and half of K should be applied in three equal splits at 10-15, 30 and 50 days after transplanting as ring method under moist soil condition and mixed thoroughly with the soil.

CHINASAK (*Brassica chinensis*)

(Var. BARI Chinasak-1)

and

BATISAK (*Brassica parachinensis*)

(Var. BARI Batisak-1)

Yield Goal: 50.0 ± 5.0 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)							OF (t/ha)
	N	P	K	S	Mg	Zn	B	
Optimum	0-38	0-11	0-15	0-4	0-1.9	-	-	5
Medium	39-76	12-22	16-30	5-8	2.0-3.8	0-1.4	0-0.8	
Low	77-114	23-33	31-45	9-12	3.9-5.7	1.5-2.8	0.9-1.6	
Very low	115-152	34-44	46-60	13-16	5.8-7.6	2.9-4.2	1.7-2.4	

Method of application:

- Half of N, K and OF and all of P and S should be applied as basal during land preparation. Remaining half of OF should be applied in pit before planting of seedlings.
- Remaining half for both N and K should be applied at 20-25 DAS under moist soil condition and mixed thoroughly with the soil.

KANGKONG (*Ipomoea aquatica*)

(Var. BARI Gima Kalmi-1)

Yield Goal: 30.0 ± 3.0 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)							OF (t/ha)
	N	P	K	S	Mg	Zn	B	
Optimum	0-30	0-10	0-13	0-5	0-1.9	-	-	5
Medium	31-60	11-20	14-26	6-10	2.0-3.8	0-1.4	0-0.8	
Low	61-90	21-30	27-39	11-15	3.9-5.7	1.5-2.8	0.9-1.6	
Very low	91-120	31-40	40-52	16-20	5.8-7.6	2.9-4.2	1.7-2.4	

Method of application:

- All of OF, P, K and S and one-third of N should be applied as basal during final land preparation.
- Remaining N should be applied as top dress after each harvest under moist soil condition and mixed thoroughly with the soil.

INDIAN SPINACH (*Basella alba*)

(Var. BARI Puisak-1, BARI Puisak-2)

Yield Goal: 30.0± 3.0 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)					OF (t/ha)
	N	P	K	S	Zn	
Optimum	0-30	0-10	0-13	0-5	-	5
Medium	31-60	11-20	14-26	6-10	0-1.0	
Low	61-90	21-30	27-39	11-15	1.1-2.0	
Very low	91-120	31-40	40-52	16-20	2.1-3.0	

Method of application:

- All of OF, P, K, S and one-third of N and K should be applied as basal during final land preparation.
- Remaining N and K should be applied in two equal splits at 14 and 28 days after planting under moist soil condition and mixed thoroughly with the soil.

SPINACH (*Spinacea oleracea*)

(Var. BARI Palongsak-1, BARI Palongsak-2)

Yield Goal: 30.0± 3.0 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)					OF (t/ha)
	N	P	K	S	Zn	
Optimum	0-20	0-8	0-10	0-4	-	5
Medium	21-40	9-16	11-20	5-8	0-1.0	
Low	41-60	17-24	21-30	9-12	1.1-2.0	
Very low	61-80	25-32	31-40	13-16	2.1-3.0	

Method of application:

- All of OF, P, K and S should be applied as basal during final land preparation.
- Nitrogen should be applied in two instalments at 21 and 42 DAS.

AMARANTHUS (*Amaranthus lividus*)

(Var. BARI Danta-1, 2, 3)

Yield Goal: 30.0± 3.0 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)					OF (t/ha)
	N	P	K	S	Zn	
Optimum	0–20	0-7	0-12	0-4	-	5
Medium	21–40	8-14	13-24	5-8	0-1.0	
Low	41–60	15-21	25-36	9-12	1.1-2.0	
Very low	61–80	22-28	37-48	13-16	2.1-3.0	

Method of application:

- All of OF, P, K and S; and one-third of N should be applied as basal during final land preparation.
- Remaining N should be applied in two equal splits at 15 and 35 DAS under moist soil condition and mixed thoroughly with the soil.

Sabuj Data shak

(Var: BARI Sabuj Dantasak-1)

Yield Goal: 30.0 ± 3.0 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)					OF (t/ha)
	N	P	K	S	Zn	
Optimum	0–20	0-7	0-12	0-4	-	5
Medium	21–40	8-14	13-24	5-8	0-1.0	
Low	41–60	15-21	25-36	9-12	1.1-2.0	
Very low	61–80	22-28	37-48	13-16	2.1-3.0	

Method of application:

- All of OF, P, K, S and half of N should be applied as basal during final land preparation.
- Remaining N should be applied as topdress at 10–15 DAS under moist soil condition.

RED AMARANTH (*Amaranthus gangeticus*)

(Var. BARI Lalshak-1)

Yield Goal: 30.0 ± 3.0 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)					OF (t/ha)
	N	P	K	S	Zn	
Optimum	0-25	0-5	0-7	0-2	-	5
Medium	26-50	6-10	8-14	3-4	0-1.0	
Low	51-75	11-15	15-21	5-6	1.1-2.0	
Very low	76-100	16-20	11-28	7-8	2.1-3.0	

Method of application:

- All of OF, P, K and S; and half of N should be applied as basal during final land preparation.
- Remaining N should be applied as topdress at 10-15 DAS under moist soil condition.

TOMATO (*Solanum lycopersicum*) (Winter)

(Var. BARI Tomato-11, BARI Tomato-14, BARI Tomato-15, BARI Tomato-16, BARI Tomato-17, BARI Tomato-18, BARI Tomato-19, BARI Tomato-20, BARI Tomato-21; Binatomato-5, Binatomato-6, Binatomato-7, Binatomato-8, Binatomato-9, Binatomato-10, Binatomato-11, Binatomato-12)

Yield Goal: 75.0 ± 7.5 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)							OF (t/ha)
	N	P	K	S	Mg	Zn	B	
Optimum	0-40	0-15	0-25	0-7	0-4	-	-	5
Medium	41-80	16-30	26-50	8-14	5-8	0-1.6	0-1.0	
Low	81-120	31-45	51-75	15-21	9-12	1.7-3.2	1.1-2.0	
Very low	121-160	46-60	76-100	22-28	13-16	3.3-4.8	2.1-3.0	

Method of application:

- Half of OF and all of P, S, Zn and B should be applied as basal during final land preparation. Remaining OF should be applied in pits before planting of seedlings.
- N and K should be applied in two equal splits at 15 and 35 DAT as ring method under moist soil condition and mixed thoroughly with the soil as soon as possible for better utilization.

TOMATO (*Solanum lycopersicum*) (Summer)

(Var. BARI Hybrid Tomato-4, BARI Hybrid Tomato-8, BARI Hybrid Tomato-10, BARI Hybrid Tomato-11, Binatomato-2, Binatomato-3)

Yield Goal: 45.0 ± 4.5 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)						OF (t/ha)
	N	P	K	S	Zn	B	
Optimum	0-30	0-11	0-25	0-7	-	-	4
Medium	31-60	12-22	26-50	8-14	0-1.5	0-1.0	
Low	61-90	23-33	51-75	15-21	2.3-3.0	1.1-2.0	
Very low	91-120	34-44	76-100	22-28	4.5-4.5	2.1-3.0	

Method of application:

- a) Half of OF and all of P, S, Zn and B should be applied as basal during final land preparation. Remaining OF should be applied in pits before planting of seedlings.
- b) N and K should be applied in three equal splits at 20, 40 and 60 DAT as ring method under moist soil condition and mixed thoroughly with the soil as soon as possible for better utilization.

BRINJAL (*Solanum melongena*)

(Var. BARI Begun-4, BARI Begun-5, BARI Begun-6, BARI Begun-8, BARI Begun-9, BARI Begun-10, BARI Begun-11, BARI Begun-12; BARI Hybrid Begun-2, BARI Hybrid Begun-3, BARI Hybrid Begun-4, BARI Hybrid Begun-6; BARI Bt Begun-1, BARI Bt Begun-2, BARI Bt Begun-3, BARI Bt Begun-4)

Yield Goal: 50.0 ±5.0 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)							OF (t/ha)
	N	P	K	S	Mg	Zn	B	
Optimum	0-45	0-15	0-40	0-5	0-4	-	-	7
Medium	46-90	21-30	41-80	6-10	5-8	0-1.5	0-1.0	
Low	91-135	41-45	81-120	11-15	9-12	1.6-3.0	1.1-2.0	
Very low	136-180	61-60	121-160	16-20	13-16	3.1-4.5	2.1-3.0	

Method of application:

- a) Half of OF and all of P, S, Zn and B should be applied as basal during final land preparation. Remaining OF should be applied in pits before planting of seedlings.
- b) N and K should be applied in three equal splits at 20, 40 and 60 DAT as ring method under moist soil condition and mixed thoroughly with the soil as soon as possible for better utilization.
- c) Instead of urea, Urea Supper Granule (USG) might be applied 7–8 cm deep into the soil and 8–10 cm apart from the plant and as ring method.

OKRA (*Abelmoschus esculentus*)

(Var. BARI Dherosh-2)

Yield Goal: 25.0 ± 2.5 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)						FYM* (t/ha)
	N	P	K	S	Zn	B	
Optimum	0–30	0–10	0–23	0–5	–	–	5
Medium	31–60	11–20	24–46	6–10	0.0–1.0	0.0–0.7	
Low	61–90	21–30	47–69	11–15	1.1–2.0	0.8–1.4	
Very low	91–120	31–40	70–92	16–20	2.1–3.0	1.5–2.1	

Method of application:

- All of OF and all of P, S, Zn and B; and one-fourth of N should be applied as basal during final land preparation.
- Remaining three-fourth N should be applied at 20, 40 and 60 DAS under moist soil condition and mixed thoroughly with the soil.

RADISH (*Raphanus sativus*)

(Var. BARI Mula-1, BARI Mula-2, BARI Mula-3, BARI Mula-4)

Yield Goal: 60.0 ± 6.0 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)					OF (t/ha)
	N	P	K	S	Zn	
Optimum	0–37	0–15	0–30	0–6	–	5
Medium	38–74	16–30	31–60	7–12	0–1.5	
Low	75–111	31–45	61–90	13–18	1.6–3.0	
Very low	112–148	46–60	91–120	19–24	3.1–4.5	

Method of application:

- All of OF and P, S, Zn; and one-third of N and K should be applied as basal during final land preparation.
- Remaining N and K should be side dressed in two equal splits at 21 and 35 DAS under moist soil condition and mixed thoroughly with the soil.

CARROT (*Daucus carota*)

(All cultivars)

Yield Goal: 25 ± 2.5 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)					OF (t/ha)
	N	P	K	S	Zn	
Optimum	0–33	0–10	0–30	0–5	–	3
Medium	34–66	11–20	31–60	6–10	0–1.0	
Low	67–99	21–30	61–90	11–15	1.1–2.0	
Very low	100–132	31–40	91–120	16–20	2.1–3.0	

Method of application:

- All of OF, P, K, S, Zn and one-third of N should be applied as basal during final land preparation.
- Remaining N should be applied in two equal splits at the 3rd and 5th weeks of sowing under moist soil condition and mixed thoroughly with the soil.

COUNTRY BEAN (*Dolichos lablab*)

(Var. BARI Sheem-1, BARI Sheem-2, BARI Sheem-4, BARI Sheem-5, BARI Sheem-6, BARI Sheem-7, BARI Sheem-8, BARI Sheem-9, BARI Sheem-10, BARI Jack Sheem-1, IPSA Shim-2)

Yield Goal: 25.0 ±2.5 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)							OF (t/ha)
	N	P	K	S	Mg	Zn	B	
Optimum	0–15	0–12	0–23	0–4	0–2.5	–	–	5
Medium	16–30	13–24	24–46	5–8	2.6–5.0	0–1.5	0–1.0	
Low	31–45	25–36	47–69	9–12	5.1–7.5	2.2–3.0	1.1–2.0	
Very low	46–60	37–48	70–92	13–16	7.6–10	4.3–4.5	2.1–3.0	

Method of application:

- All of OF, P, K, S, Mg, Zn and B; and half of N should be applied as basal during final land preparation.
- Remaining half of N should be applied during 25–30 DAT under moist soil condition and mixed thoroughly with the soil.

YARDLONG BEAN (*Vigna unguiculata*)

(Var. BARI Barboti-1, BARI Barboti-2, Kagor Natoki)

Yield Goal: 15.0 ±1.5 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)						OF (t/ha)
	N	P	K	S	Zn	B	
Optimum	0-12	0-7	0-14	0-5	–	–	5
Medium	13-24	8-14	15-28	6-10	0.0–1	0.0–0.7	
Low	25-36	15-21	29-42	11-15	0.7–2	0.7–1.4	
Very low	37-48	22-28	43-56	16-20	1.3– 3.0	1.3–2.1	

Method of application:

- All fertilizers except N should be applied as basal during final land preparation.
- Nitrogen should be applied in two equal splits during 2nd and 4th weeks of germination.
- Rhizobium* inoculum (at the rate of 50 g/kg seed) should be used if available and in that case N fertilizer should not be used.

FRENCH BEAN (*Phaseolus vulgaris*)

(Var. BARI Jharsheem-1, BARI Jharsheem-2, BARI Jharsheem-3)

Yield Goal: 15.0 ± 1.5 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)						OF (t/ha)
	N	P	K	S	Zn	B	
Optimum	0-19	0-10	0-20	0-4	–	–	3
Medium	20-38	11-20	21-40	5-8	0.0–1.0	0.0–0.7	
Low	39-57	21-30	41-60	9-12	1.1–2.0	0.8–1.4	
Very low	58-76	31-40	61-80	13-16	2.1–3.0	1.5–2.1	

Method of application:

- All of OF, P, K, S, Zn and B; and one-third of N should be applied as basal during final land preparation.
- Remaining N should be side dressed at 20 and 35 DAS under moist soil condition and mixed thoroughly with the soil.

GARDEN PEA (*Pisum sativum*)

(Var. BARI Motoshuti-1, BARI Motoshuti-2, BARI Motoshuti-3, IPSA Motorshuti-1, IPSA Motorshuti-2, IPSA Motorshuti-3)

Yield Goal: 15.0 ±1.5 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)						OF (t/ha)
	N	P	K	S	Zn	B	
Optimum	0-10	0-6	0-13	0-4	—	—	5
Medium	11-20	7-12	14-26	5-8	0.0-1	0.0-0.7	
Low	21-30	13-18	27-39	9-12	0.7-2	0.7-1.4	
Very low	31-40	19-24	40-52	13-16	1.3- 3.0	1.3-2.1	

Method of application:

- All of OF, P, K, S, Zn, B and one-third of N should be applied as basal during final land preparation.
- Remaining N should be side dressed at 20 and 35 DAS under moist soil condition and mixed thoroughly with the soil.

SWEET GOURD (*Cucurbita moschata*)

(Var. BARI Mistikumra -1, BARI Mistikumra -2, BARI Mistikumra -3; BARI Hybrid Mistikumra-1, BARI Hybrid Mistikumra-2, BARI Hybrid Mistikumra-3)

Yield Goal: 45.0 ±4.5 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)							OF (t/ha)
	N	P	K	S	Mg	Zn	B	
Optimum	0-28	0-14	0-28	0-7	0-2.4	-	-	5
Medium	29-56	15-28	31-56	8-14	2.5-4.8	0-1.9	0-1.0	
Low	56-84	29-42	61-84	15-21	4.9-7.2	2.0-3.8	1.1-2.0	
Very low	85-92	43-56	91-112	22-28	7.3-9.6	3.9-5.7	2.1-3.0	

Method of application:

- All of OF, P, K, S, Zn and B should be applied in pits 5–7 days before planting and mixed thoroughly with the soil.
- Nitrogen should be applied around the plant as side dressing at 15, 35, 55 and 75 DAT under moist soil condition and mixed thoroughly with the soil.

BOTTLE GOURD (*Lagenaria siceraria*)

(Var. BARI Lau-1, BARI Lau-2, BARI Lau-3, BARI Lau-4, BARI Lau-5 and BARI Seeta Lau-1)
and

ASH GOURD (*Benincasa hispida*)

(Var. BARI Chal Kumra-1, IPSA Ash gourd-1)

Yield Goal: 60.0 ± 6.0 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)							OF (t/ha)
	N	P	K	S	Mg	Zn	B	
Optimum	0-33	0-13	0-20	0-7	0-3	-	-	5
Medium	34-66	14-26	21-40	8-14	4-6	0-1.5	0-1.0	
Low	67-99	27-39	41-60	15-21	7-9	2.3-3.0	1.1-2.0	
Very low	100-132	40-52	61-80	22-28	10-12	4.5-4.5	2.1-3.0	

Method of application:

- All of OF, P, K, S, Mg, Zn and B should be applied in pit 5–7 days before planting and mixed thoroughly with the soil.
- Nitrogen should be applied around the plant as side dressing at 15, 35, 55 and 75 DAT under moist soil condition and mixed thoroughly with the soil.

TEASLE GOURD (*Momordica cochinchinensis*)

(KAKROL) (All cultivars)

Yield Goal: 25.0 ±2.5 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)							OF (t/ha)
	N	P	K	S	Mg	Zn	B	
Optimum	0–20	0-9	0-13	0-5	0-2.5	-	-	5
Medium	21–40	12-18	14-26	6-10	2.6-5.0	1.1-1.5	0-1.0	
Low	41–60	23-27	27-39	11-15	5.1-7.5	1.6-2.0	1.1-3.0	
Very low	61–80	34-36	40-52	16-20	7.6-10	2.1-4.5	2.1-3.0	

Method of application:

- All of OF, P, K, S, Mg, Zn and B should be applied in pits 5–7 days before planting and mixed thoroughly with the soil.
- Nitrogen should be applied around the plant as side dressing at 15, 35, 55 and 75 DAT under moist soil condition and mixed thoroughly with the soil as soon as possible for better utilization.

BITTER GOURD (*Momordica charantia*)

(Var. BARI Karolla-1, BARI Karolla-2, BARI Karolla-3, BARI Karolla-4; BARI Hybrid Karolla-1, BARI Hybrid Karolla-2, BARI Hybrid Karolla-3)

Yield Goal: 25 ± 2.5 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)							OF (t/ha)
	N	P	K	S	Mg	Zn	B	
Optimum	0-25	0-14	0-30	0-5	0-2.5	-	-	5
Medium	26-50	15-28	31-60	6-10	2.6-5.0	0-1.5	0-1.0	
Low	51-75	29-42	61-90	11-15	5.1-7.5	2.3-3.0	1.1-2.0	
Very low	76-100	43-56	91-120	16-20	7.6-10	4.5-4.5	2.1-3.0	

Method of application:

- All of OF, P, K, S, Mg, Zn and B should be applied in pits 5–7 days before planting and mixed thoroughly with the soil.
- Nitrogen should be applied around the plant as side dressing at 15, 35, 55 and 75 DAT under moist soil condition and mixed thoroughly with the soil.

POINTED GOURD (*Trichosanthes dioica*)

(Var. BARI Potal-1, BARI Potal-2; BARI Hybrid Potal-1)

Yield Goal: 25.0 ±2.5 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)							OF (t/ha)
	N	P	K	S	Mg	Zn	B	
Optimum	0–30	0-9	0-13	0-5	0-2.5	-	-	5
Medium	31–60	10-18	14-26	6-10	2.6-5.0	0-1.5	0-0.8	
Low	61–90	19-27	27-39	11-15	5.1-7.5	1.6-3.0	0.9-1.6	
Very low	91–120	28-36	40-52	16-20	7.6-10	3.1-4.5	1.7-2.4	

Method of application:

- All of OF, P, K, S, Mg, Zn and B should be applied in pits 5-7 days before planting and mixed thoroughly with the soil.
- Nitrogen should be applied around the plant as side dressing at 20, 40 and 60 DAT planting and mixed thoroughly with the soil.

SNAKE GOURD (*Trichosanthes anguina*)

(Var: BARI Chicinga-1 and others),

RIDGE GOURD (*Luffa acutangula*)

(Var: BARI Jhinga-1, BARI Jhinga-2 and others)

and

SPONGE GOURD (*Luffa cylindrica*)

(Var: BARI Dhundul-1, BARI Dhundul-2, BARI Hybrid Dhundul-1 and others)

Yield Goal: 35.0 ±3.5 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)							OF (t/ha)
	N	P	K	S	Mg	Zn	B	
Optimum	0–25	0–12	0–20	0–7	0–2.4	-	-	5
Medium	26–50	13–24	21–40	7–14	2.5–4.8	0–2.2	0–1.0	
Low	51–75	25–36	41–60	13–21	4.9–7.2	2.3–4.4	1.1–2.0	
Very low	76–100	37–48	61–80	19–28	7.3–9.6	4.5–4.8	2.1–3.0	

Method of application:

- All of OF, P, K, S, Mg, Zn and B should be applied in pits 5–7 days before planting and mixed thoroughly with the soil.
- Nitrogen should be applied around the plant as side dressing at 15, 35, 55 and 75 DAT under moist soil condition and mixed thoroughly with the soil.

CUCUMBER (*Cucumis sativus*)

(Var: All cultivars)

Yield Goal: 25 ± 2.5 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)						OF (t/ha)
	N	P	K	S	Zn	B	
Optimum	0–25	0–9	0–15	0–5	-	-	5
Medium	26–50	10–18	16–30	6–10	0–2.2	0–1.0	
Low	51–75	19–27	31–45	11–15	2.3–4.4	1.1–2.0	
Very low	76–100	28–36	46–60	16–20	4.5–4.8	2.1–3.0	

Method of application:

- All of OF, P, S, Zn and B; and one-third of N and K should be applied in pit 5–7 days before planting and mixed thoroughly with the soil.
- Remaining N and K should be applied during 20, 35 and 50 DAT around the plant as side dressing under moist soil condition and mixed thoroughly with the soil.

SQUASH

(Var. OP)

Yield Goal: 30.0 ±3.0 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)						OF (t/ha)
	N	P	K	S	Zn	B	
Optimum	0-25	0-9	0-20	0-5	–	–	5
Medium	26-50	10-18	21-40	6-10	0.0–1.0	0.0–0.7	
Low	51-75	19-27	41-60	11-15	1.1–2.0	0.8–1.4	
Very low	76-100	28-36	61-80	16-20	2.1–3.0	1.5–2.1	

Method of application:

- a) All of OF, P, K, S, Zn and B should be applied in pit 5–7 days before planting and mixed thoroughly with the soil.
- b) N should be applied around the plant as side dressing at 15, 35, 55 and 75 DAT under moist soil condition and mixed thoroughly with the soil.

7.7 Spice Crops

ONION (*Allium cepa*)

(Var. BARI Peyaz-1, BARI Peyaz-4 and BARI Peyaz-5, BARI Peyaz-6, BARI Peyaz-7)

Yield Goal: 20.0 ± 2.0 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)						OF (t/ha)
	N		K	S	Zn	B	
Optimum	0-30	0-10	0-25	0–10	-	–	5
Medium	31-60	11-20	26-50	11–20	0-1.3	0.0–0.7	
Low	61-90	21-30	51-75	21–30	1.6-2.6	0.8–1.4	
Very low	91-120	31-40	76-100	31–40	2.7-3.9	1.5–2.1	

Method of application:

- a) All of OF, P, S, Zn and B and half of N and K should be applied as basal during final land preparation.
- a) Remaining N and K should be applied in 2 equal splits at 25 and 50 DAP for bulb production; and in 3 equal splits at 25, 50 and 75 days after planting for seed production.

SUMMER ONION (*Allium cepa*)

(Var. BARI Peyaz-2, BARI Peyaz-3, BARI Peyaz-5)

Yield Goal: 12.0 ± 1.2 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)						OF (t/ha)
	N	P	K	S	Zn	B	
Optimum	0–25	0–10	0–20	0–8	–	–	3
Medium	26–50	11–20	21–40	9–16	0–1.0	0.0–0.5	
Low	51–75	21–30	41–60	17–24	1.1–2.0	0.6–1.0	
Very low	76–100	31–40	61–80	25–32	2.1–3.0	1.1–1.5	

Method of application:

- b) All of OF, P, S, Zn and B and half of N and K should be applied as basal during final land preparation.
- b) Remaining N and K should be applied in 2 equal splits at 25 and 50 DAP for bulb production; and in 3 equal splits at 25, 50 and 75 days after planting for seed production.

GARLIC (*Allium sativum*)

(Var. BARI Rashun-1, BARI Rashun-2, BARI Rashun-3, BARI Rashun-4, BAU Rashun-2
BAU Rashun-2)

Yield Goal: 15.0 ± 1.5 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)						OF (t/ha)
	N	P	K	S	Zn	B	
Optimum	0-30	0-10	0-25	0–10	-	-	5
Medium	31-60	11-20	26-50	11–20	0-1.3	0-0.7	
Low	61-90	21-30	51-75	21–30	1.4-2.6	0.8-1.4	
Very low	91-120	31-40	76-100	31–40	2.6-3.9	1.5-2.1	

Method of application:

- a) All of OF, P, S, Zn and B, and half of N and K should be applied as basal during final land preparation.
- b) Remaining N and K should be applied at the time of first mulching under moist soil condition and mixed thoroughly with the soil as soon as possible for better utilization.

GARLIC (*Allium sativum*)

(Var. BARI Rashun-1)

(under zero tillage)

Yield Goal: 13.0 ± 1.3 t/ha

Nutrient Recommendation (kg/ha)				
N	P	K	S	Zn
160	50	125	25	3

Method of application:

- Under zero tillage, garlic is cultivated widely in Chalan Beel areas of Natore, Sirajganj and Pabna districts after harvesting of deep water aman rice. Before cultivation of garlic, rice straw should be removed from the land. Then all of OF, P, K, S and Zn; and one-third of N should be applied as broadcast on muddy soils. After application of fertilizers garlic should be planted by dibbling shallowly (1/3rd of the clove into the soil) and the land should be covered again with the rice straw.
- Remaining N should be applied in two equal splits - the first split during 25-30 DAP and the second split during 55-60 DAP. The N should be applied by topdressing on the straw mulch followed by gentle shaking of the straw for allowing the fertilizer to reach on to the soil surface.
- After 2-3 hours of N topdressing light irrigation should be provided. Care should be taken so that there is no water logging after irrigation to avoid damage of the crop.

GINGER (*Zingiber officinale*)

(Var. BARI Ada-1, BARI Ada-2, BARI Ada-3)

Yield Goal: 30.0 ± 3.0 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)						OF (t/ha)
	N	P	K	S	Zn	B	
Optimum	0-45	0-15	0-40	0-5	—	—	3
Medium	46-90	16-30	41-80	6-10	0.0-1.5	0.0-0.7	
Low	91-135	31-45	81-120	11-15	1.6-3.0	0.8-1.4	
Very low	136-180	46-60	121-160	16-20	3.1-4.5	1.5-2.1	

Method of application:

- All of OF, P, K, S, Zn and B, and half of N should be applied as basal during final land preparation.
- Remaining N should be applied at the 6th week of planting under moist soil condition and mixed thoroughly with the soil as soon as possible for better utilization.

TURMERIC (*Curcuma longa*)

(Var. BARI Halud-1 (Sinduri), BARI Halud-2 (Dimla) and BARI Halud-3, BARI Halud-3, BARI Halud-4, BARI Halud-5)

ield Goal: 30.0 ± 3.0 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)						OF (t/ha)
	N	P	K	S	Zn	B	
Optimum	0–35	0–10	0–40	0–5	–	–	3
Medium	36–70	11–20	41–80	6–10	0.0–1.5	0.0–0.7	
Low	71–105	21–30	81–120	11–15	1.6–3.0	0.8–1.4	
Very low	106–140	31–40	121–160	16–20	3.1–4.5	1.5–2.1	

Method of application:

- a) All of OF, P, K, S, Zn and B, and half of N should be applied as basal during final land preparation.
- b) Remaining N should be applied in two equal splits at 80 and 110 DAP under moist soil condition and mixed thoroughly with the soil as soon as possible for better utilization.

CHILLI (*Capsicum annum*)

(Var. BARI Morich-1, BARI Morich-2, BARI Morich-4, BARI Morich-5, BARI Morich-6, Bogra Bogra local, Jamalpur local)

Yield Goal: 2.5± 0.25 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)						OF (t/ha)
	N	P	K	S	Zn	B	
Optimum	0-25	0-12	0-30	0-4	-	–	3
Medium	26-50	13-24	31-60	5-8	0-1.0	0.0–0.7	
Low	51-75	25-36	61-90	9-12	1.1-2.0	0.8–1.4	
Very low	76-100	37-48	91-120	11-16	2.1-3.0	1.5–2.1	

Method of application:

- a) All of OF, P, K, S, Zn and B; and half of N should be applied as basal during final land preparation.
- b) Remaining N should be applied in three equal splits at 25, 50 and 70 DAP under moist soil condition and mixed thoroughly with the soil.

SWEET PEPPER (*Capsicum*) (*Capsicum annuum* L.)

(Var: BARI Misty Morich-1)

Yield Goal: 20.0 ± 2.0 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)						OF (t/ha)
	N	P	K	S	Zn	B	
Optimum	0-35	0-12	0-33	0-6	-	-	5
Medium	36-70	13-24	34-66	7-12	0-1.0	0-0.7	
Low	71-105	25-36	67-99	13-18	1.1-2.0	0.8-1.4	
Very low	106-140	37-48	100-132	19-24	2.1-3.0	1.5-2.1	

Method of application:

- c) All of OF, P, K, S, Zn and B; and half of N should be applied as basal during final land preparation.
- d) Remaining N should be applied in three equal splits at 25, 50 and 70 DAP under moist soil condition and mixed thoroughly with the soil.

CORIANDER (*Coriandrum sativum*)

(Var. BARI Dhonia-1, BARI Dhonia-2)

Yield Goal: 1.5 ± 0.15 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)						OF (t/ha)
	N	P	K	S	Zn	B	
Optimum	0-20	0-7	0-18	0-5	-	-	3
Medium	21-40	8-14	19-36	6-10	0.0-1.3	0.0-1.0	
Low	41-60	15-21	37-54	11-15	1.4-2.6	1.1-2.0	
Very low	61-80	22-28	55-72	16-20	2.7-3.9	2.1-3.0	

Method of application:

- a) All of OF, P, K, S, Zn and B, and half of N should be applied as basal during final land preparation.
- b) Remaining N should be applied as top dress at 30 DAS under moist soil condition and mixed thoroughly with the soil.

BLACK CUMIN (*Nigella sativa*)

(Var. BARI Kalozira-1)

Yield Goal: 1.0 ± 0.10 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)						OF (t/ha)
	N	P	K	S	Zn	B	
Optimum	0–20	0–8	0–15	0–5	–	–	2
Medium	21–40	9–16	16–30	6–10	0–1.0	0.0–0.7	
Low	41–60	17–24	31–45	11–15	1.1–2.0	0.8–1.4	
Very low	61–80	25–32	46–60	16–20	2.1–3.0	1.5–2.1	

Method of application:

- All of OF, P, K, S, Zn and B should be applied as basal during final land preparation.
- Nitrogen should be applied in two equal splits as topdress at 30 and 55 DAS under moist soil condition.

FENUGREEK (*Trigonella foenum-graecum*)

(Var. BARI Methi-1, BARI Methi-2, BARI Methi-3)

Yield Goal: 2.0 ± 0.2 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)						OF (t/ha)
	N	P	K	S	Zn	B	
Optimum	0–25	0–10	0–20	0–5	–	–	2
Medium	26–50	11–20	21–40	6–10	0–1.0	0.0–0.7	
Low	51–75	21–30	41–60	11–15	1.1–2.0	0.8–1.4	
Very low	76–100	31–40	61–80	16–20	2.1–3.0	1.5–2.1	

Method of application:

- All of OF, P, K, S, Zn and B should be applied as basal during final land preparation.
- Nitrogen should be applied in two equal splits as topdress at 30 and 55 DAS under moist soil condition.

BLACK PEPPER (*Piper nigrum*)

(Var. BARI Golmorich-1, Jainta Gulmorich)

Yield Goal: 4–5 kg/plant/year

Age of plant (Year)	Nutrient Recommendation (g/plant/year)			OF (kg/plant)
	N	P	K	
Before planting (in pit)	–	50	50	2-3
1	25	0	0	-
2	50	30	50	-
≥ 3	75	40	75	–

*S, Zn and B fertilizers to be sprayed if needed based on deficiency symptoms.

Method of application:

- Before planting, recommended OF, P and K should be applied into the pits 10–15 days before planting of seedling and mixed thoroughly with the soil followed by irrigation.
- For growing plant N, P and K should be applied in two equal splits during April–May and September–October under moist soil condition and mixed thoroughly with the soil followed by irrigation.

7.8 Fruit Crops

MANGO (*Mangifera indica*)

(Var. BARI Aam-3, BARI Aam-4, BARI Aam-8, BARI Aam-9, BARI Aam-10, BARI Aam-11, BARI Aam-12, BARI Aam-13, BARI Aam-14, BARI Aam-15, BARI Aam-16, BARI Aam-17, BARI Aam-18)

Yield Goal: 350± 35 kg/tree (above 20 years old tree)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)						OF (kg/tree/year)
	N	P	K	S	Zn	B	
Before planting (in Pit)	-	100	100	50	10	10	10
0–1	100	30	50	20	5	5	10
2–4	210	50	90	30	5	5	15
5–7	390	80	120	50	5	5	20
8–10	500	100	150	70	8	8	25
11–15	550	150	200	80	8	8	30
16–20	800	200	230	100	10	10	35
>20	1000	300	300	130	10	12	40

Method of application:

- 1) Before 15-20 days of sapling plantation, full amount of recommended organic and chemical fertilizers should be applied and mixed thoroughly with the soil into the pits followed by irrigation.
- 2) After plantation, fertilizers and manure should be applied in two equal splits. One half should be applied just after harvesting of fruits (June-July) and another half should be applied in September-October. Fertilizer should be applied around the tree up to canopy spread leaving 1.0-1.5 m from the tree base and mixed thoroughly with the soil followed by irrigation.

MANGO (*Mangifera indica*)

(Var. BAU Aam-1, BAU Aam-2, BAU Aam-3, BAU Aam-4, BAU Aam-5, BAU, Aam-6, BAU Aam-11, BAU Aam-14, BAU Aam-15)

Yield Goal: 300± 30 kg/tree (above 20 years old tree)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)						OF (kg/tree/year)
	N	P	K	S	Zn	B	
Before planting (in Pit)	-	200	-	111	-	41	15
0–1	130	100	120	-	7	-	5
2–3	261	200	200	33	9	-	6
4–5	370	300	240	67	17	18	7
6–7	609	500	360	133	17	18	8
8–10	848	700	480	200	26	29	9
11–15	1087	900	600	267	26	29	10
16–20	1326	1150	720	356	35	41	11
>20	1565	1400	800	444	44	53	12

Method of application: Same as BARI varieties

JACKFRUIT (*Artocarpus heterophyllus*)

(Var: BARI Kanthal-1, BARI Kanthal-2, BARI Kanthal-3, BARI Kanthal-4, BARI Kanthal-5, BARI Kanthal-6, BAU Kanthal-1)

Yield goal: 1000 ±100 kg/tree/year (above 20 years old tree)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)						OF (kg/tree/year)
	N	P	K	S	Zn	B	
Before planting (in Pit)	-	100	125	35	-	-	25
0–1	150	75	150	20	2	-	-
2–4	250	100	250	30	4	-	15
5–7	350	150	350	35	4	2	20
8–10	450	200	450	45	6	4	25
11–15	600	280	550	60	7	5	30
16–20	750	300	650	90	8	8	35
>20	800	400	750	100	10	10	40

Method of application: Same as mango.

BANANA (*Musa paradisiaca*)

(Var. BARI Kola-1, BARI Kola-2, BARI Kola-3, BARI Kola-4)

Yield Goal: 50.0± 5.0 t/ha

Method of application	Nutrient Recommendation (g/plant)						OF (kg/pit)
	N	P	K	S	Zn	B	
Before planting (pit)	0	100	0	50	3	3	15
1st TD*	50	0	70	0	0	0	0
2nd TD	50	0	70	0	0	0	0
3rd TD	50	0	70	0	0	0	0
4th TD	100	0	130	0	0	0	0
Total	250	100	340	50	3	3	15

*TD: Top dressing

Method of application:

- a) Before 15-20 days of sapling plantation, whole amount of recommended organic and chemical fertilizers should be applied and mixed thoroughly with the soil in the pit followed by irrigation.
- b) The N and K should be applied in four equal splits starting after two months of sucker planting. The second and third TD should be done at two month-intervals and the last TD should be applied after emergence of inflorescence.

PAPAYA (*Carica papaya*)

(Var. BARI papaya-1 (Shahi Pepe))

Yield Goal: 50.0 ± 5.0 t/ha

Method of application	Nutrient Recommendation (g/plant)						OF (kg/pit)
	N	P	K	S	Zn	B	
Before planting (pit)	0	120	0	60	3	5	15
1st TD	35	0	35	0	0	0	0
2nd TD	35	0	35	0	0	0	0
3rd TD	40	0	40	0	0	0	0
4th TD	60	0	60	0	0	0	0
5th TD	60	0	60	0	0	0	0
6th TD	50	0	50	0	0	0	0
Total	280	120	280	60	3	5	15

Method of application:

- a) Before 15-20 days of sapling plantation, the full amount of recommended organic and chemical fertilizers should be applied and mixed thoroughly with the soil into the pits followed by irrigation.
- b) The 1st top dressing (TD) of N and K should be done around the plant after 30 days of sapling transplantation and mixed thoroughly with the soil followed by irrigation.
- c) The remaining N and K should be applied at 30 day- intervals as the 2nd, 3rd, 4th, 5th and 6th TD until two months before final harvest.

GUAVA (*Psidium guajava*)

(Var: Kazi payara, BARI Payara-2, BARI Payara-3, BARI Payara-4; BAU Payara-7, BAU Payara-8, BAU Payara-9)

Yield Goal: 75 ± 5 kg/tree/year (above 10 years old tree)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)						OF (kg/tree/year)
	N	P	K	S	Zn	B	
Before planting (pit)	-	60	125	10	5	2	15
0-1	50	70	100	15	5	2	10
2-4	160	80	250	20	8	5	10
5-7	250	120	300	25	8	8	15
8-10	300	150	350	35	10	10	20
>10	400	200	450	40	10	10	25

Method of application

- 1) Before 15-20 days of sapling plantation, entire amount of recommended organic and chemical fertilizers should be applied and mixed thoroughly with the soil into the pits followed by irrigation.
- 2) After plantation, fertilizers and manure should be applied in two equal splits. One half should be applied just after harvesting of fruits (June-July) and another half should be applied in September-October. Fertilizer should be applied around the tree up to canopy spread leaving 1.0-1.5 m from the tree base and mixed thoroughly with the soil followed by irrigation.

COCONUT (*Cocos nucifera*)

(Var: BARI Narikel-1, BARI Narikel-2)

Yield Goal: 80 ±5 nut/tree/year (above 20 years old tree)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)						OF (kg/tree/year)
	N	P	K	S	Zn	B	
Before planting (pit)	-	60	200	45	20	5	25
0-1	100	30	200	20	10	2	10
2-4	150	50	250	30	15	3	15
5-7	220	80	350	50	20	3	20
8-10	425	100	600	55	20	5	20
11-15	550	160	950	60	25	10	25
16-20	650	160	1200	70	30	10	25
>20	800	250	1500	80	30	15	30

Method of application

- Before 15-20 days of sapling plantation, full amount of recommended organic and chemical fertilizers should be applied and mixed thoroughly with the soil into the pits followed by irrigation.
- After plantation fertilizers and manure should be applied in two equal splits. First installment should be applied during April to May and second installment should be applied during September to October. For orchards, fertilizers should be broadcast and mixed thoroughly with the soil by shallow plowing with a power tiller, followed by irrigation. For individual tree, fertilizer should be applied by drilling with a peg to a depth of 15-20 cm around the tree up to 2-3 m, leaving 1.0 m from the tree base, and mixed thoroughly with the soil followed by irrigation.

PINEAPPLE (*Ananas sativus*)

(Var. Giant kew, Honey queen, Ghorasal)

Yield Goal: 30.0 ± 3.0 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha/year)					OF (t/ha)
	N	P	K	S	Zn	
Optimum	0–60	0–25	0–60	0–7	–	3
Medium	61–120	26–50	61–120	8–14	0.0–1.5	
Low	121–180	51–75	121–180	15–21	1.6–3.0	
Very low	181–240	76–100	181–240	22–28	3.1–4.5	

Method of application:

- a) All of OF, P, S and Zn should be applied as basal during final land preparation.
- b) The N and K should be applied in five equal splits as side dressing at 1-month intervals starting from 4–5 months after planting and mixed thoroughly with the soil followed by irrigation.

LITCHI (*Litchi chinensis*)

(Var: BARI Litchu-2, BARI Litchu-3, BARI Litchu-4, BAU Litchu-3, BAU Litchu-4)

Yield Goal: 150 ± 15 kg/tree/year (above 20 years old tree)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)						OF (kg/tree/year)
	N	P	K	S	Zn	B	
Before planting (pit)	-	100	200	20	15	4	25
0-1	160	80	150	25	20	5	10
2-4	200	100	280	40	25	5	15
5-7	250	150	400	50	30	5	20
8-10	400	250	500	60	35	8	25
11-15	600	300	650	60	40	10	30
16-20	750	350	800	70	45	12	35
>20	1000	400	900	80	50	15	40

Method of application:

- 1) Before 15-20 days of sapling plantation, entire amount of OF, P, K, S, Zn and B should be applied and mixed thoroughly with the soil into the pits, followed by irrigation.
- 2) In the 1st year, OF and N should be applied at a time around the tree in September.
- 3) After plantation, fertilizers and manure should be applied in two equal splits. One half should be applied just after harvesting of fruits (June-July) and another half should be applied in September-October. Fertilizer should be applied around the tree up to canopy spread.

CITRUS

PUMMELO (*Citrus grandis*)

(Var: BARI Batabilebu-3, BARI Batabilebu-4, BARI Batabilebu-5, BARI Batabilebu-6)

Yield Goal: 120± 12 kg/tree/year (above 15 years old tree)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)						OF (kg/tree/year)
	N	P	K	S	Zn	B	
Before planting (pit)	-	60	50	45	-	-	15
0-1	100	40	90	20	5	2	10
2-4	150	50	160	25	5	2	15
5-7	250	75	250	30	20	5	20
8-10	300	100	275	40	20	6	25
11-15	350	120	350	50	25	8	30
>15	450	150	400	60	25	10	35

Method of application

- 1) Before 15-20 days of sapling plantation, full amount of recommended organic and chemical fertilizers should be applied and mixed thoroughly with the soil into the pits followed by irrigation.
- 2) After plantation fertilizers and manure should be applied in three equal splits. First installment should be applied during February, and the second and third installments during May and October, respectively. Fertilizers should be applied around the tree up to canopy spread leaving 50-100 cm from the tree base and mixed thoroughly with the soil followed by irrigation. Under hill slope conditions, drilling method should be followed.
- 3) For the highly acidic soils, dolomite should be applied at the rate of 250g/tree in every 3-4 years.

MANDARIN (*Citrus reticulanta*)

(Var: BARI Komola-2, BARI Komola-3)

Yield Goal: 50±5 kg/tree/year (above 15 years old tree)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)						OF (kg/tree/year)
	N	P	K	S	Zn	B	
Before planting (pit)	-	60	50	-	-	-	15
0-1	100	30	80	20	5	2	10
2-4	150	50	120	25	8	2	15
5-7	200	75	180	30	10	3	20
8-10	250	100	250	40	12	4	25
11-15	350	150	300	50	15	6	30
>15	500	200	400	60	15	8	35

Method of application

- 1) Before 15-20 days of sapling plantation, entire amount of recommended organic and chemical fertilizers should be applied and mixed thoroughly with the soil into the pits followed by irrigation.
- 2) After plantation fertilizers and manure should be applied in three equal splits. First installment should be applied during February, and the second and third installments should be applied during May and September, respectively. Fertilizer should be applied around the tree up to canopy spread leaving 50-100 cm from the tree base and mixed thoroughly with the soil followed by irrigation. Under hill slope conditions, drilling method should be followed.
- 3) For the highly acidic soils, dolomite should be applied at the rate of 250g/tree at every 3-4 years interval.

SWEET ORANGE (*Citrus sinensis*)

(Var. BARI Malta-1, BARI Malta-2, BAU Malta-1)

Yield Goal: 60 ± 6.0 kg/tree/year (Above 15 years old tree)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)						OF (kg/tree/year)
	N	P	K	S	Zn	B	
Before planting (pit)	-	60	50	-	-	-	20
0-1	125	40	75	20	5	2	10
2-4	200	50	90	25	8	2	15
5-7	250	75	140	40	10	5	20
8-10	320	100	175	50	12	5	30
11-15	350	150	225	50	15	8	35
>15	480	200	300	60	15	8	40

Method of application: Same as mandarin

LEMON (*Citrus limon*)

(Var. BARI Lebu-4, BARI Lebu-5, BARI Lebu-6)

Yield Goal: 20.0 ± 2.0 kg/tree/year (above 5 years old tree)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)						OF (kg/tree/year)
	N	P	K	S	Zn	B	
Before Planting (Pit)	-	50	50	-	-	-	15
0-1	125	75	75	20	5	2	10
2-4	150	90	200	30	5	2	15
5-7	250	120	250	50	8	5	20
8-10	300	160	280	80	10	6	25
>10	350	200	320	80	10	8	30

Method of application: Same as mandarin

SATKARA (*Citrus macroptera*)

(Var. BARI Satkara-1)

Yield Goal: 35.0 ± 3.5 kg/tree/year (above 10 years old tree)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)				OF (kg/tree/year)
	N	P	K	S	
Before planting (Pit)	-	50	-	-	-
0-1	100	25	75	20	5
2-4	150	35	170	25	5
5-7	200	80	250	30	8
8-10	280	100	280	30	10
>10	300	150	325	40	10

Method of application:

- 1) Before 15-20 days of sapling plantation, full amount of recommended organic and chemical fertilizers should be applied and mixed thoroughly with the soil into the pits followed by irrigation.
- 2) After plantation, fertilizers and manure should be applied in three equal splits. First installment should be applied during February, and the second and third installments should be applied during May and October, respectively. Fertilizer should be applied around the tree up to canopy spread leaving 50-100 cm from the tree base and mixed thoroughly with the soil followed by irrigation. Under hill slope conditions, drilling method should be followed.
- 3) For the highly acidic soils, dolomite should be applied at the rate of 250g/tree at every 3-4 years interval.

BER (*Zizyphus mauritiana*)

(Var. BARI Kul-1, BARI Kul-2, BARI Kul-3, BARI Kul-4, BARI Kul-5)

Yield Goal: 50±5 t/ha/year (above 20 years old tree)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)					OF (kg/tree/year)
	N	P	K	S	B	
Before planting (pit)	-	60	125	10	-	-
0-1	140	50	125	15	0	2
2-4	320	100	300	20	5	2.5
5-7	400	160	400	25	5	3
8-10	500	200	550	30	5	4
11-15	550	250	600	40	10	4
16-20	620	280	650	40	10	5
>20	750	350	750	50	10	5

Method of application

- 1) Before 15-20 days of sapling plantation, whole amount of recommended organic and chemical fertilizers should be applied and mixed thoroughly with the soil into the pits followed by irrigation.
- 2) After plantation fertilizers and manure should be applied in three equal splits. First installment should be applied during March, and the second and third installments should be applied during May and September, respectively. Fertilizer should be applied around the tree up to canopy spread leaving 1.0-1.5 m from the tree base and mixed thoroughly with the soil followed by irrigation.

GOLDEN APPLE (*Spondias pinnata*)

(Var: BARI Amra-1, BAU Amra-1, BARI Amra-2)

Yield Goal: 30± 3.0 kg/tree/year (above 10 years old tree)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)					OF (kg/tree/year)
	N	P	K	S	B	
Before planting (pit)	-	50	30	-	-	-
0-1	50	40	50	15	5	2
2-4	100	50	80	25	5	2
5-7	125	80	120	30	8	4
8-10	150	100	150	35	10	5
>10	180	125	180	50	10	5

Method of application:

- a) Before 15-20 days of sapling plantation all amount recommended organic and chemical fertilizers should be applied and mixed thoroughly with the soil into the pits followed by irrigation.
- b) After plantation fertilizers and manure should be applied in four equal splits at 2-3 months' interval during March-April, May-June, September and October. Fertilizer should be applied around the tree up to canopy spread leaving 50-100 cm from the tree base and mixed thoroughly with the soil followed by irrigation

GOLDEN APPLE (*Spondias pinnata*)**(Var. BARI Amra-2)****Yield Goal: 50± 5.0 kg/tree** (above 10 years old tree)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)					OF (kg/tree/year)
	N	P	K	S	B	
Before planting (pit)	-	75	30	-	-	-
0-1	75	50	100	20	5	2
2-4	120	75	125	30	5	2
5-7	180	100	180	40	8	4
8-10	200	120	200	50	10	5
11-15	250	140	250	60	10	8
>15	300	150	300	75	12	10

Method of application:

- a) Before 15-20 days of sapling plantation, entire amount of recommended organic and chemical fertilizers should be applied and mixed thoroughly with the soil into the pits followed by irrigation.
- b) After plantation, the fertilizers and manure should be applied in four equal splits at 2-3 months' intervals during March-April, May-June, September and October. Fertilizer should be applied around the tree up to canopy spread leaving 50-100 cm from the tree base and mixed thoroughly with the soil followed by irrigation.

SAPOTA (*Achras sapota*)

(Var. BARI Sofeda-2, BARI Sofeda-3, BAU Shofeda-1,
BAU Shofeda-2, BAU Shofeda-3, BAU Shofeda-4)

Yield Goal: 150 ± 15 kg/tree/year (above 20 years old tree)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)						OF (kg/tree/year)
	N	P	K	S	Zn	B	
Before planting (pit)	-	60	45	20	-	-	15
0-1	60	30	75	20	5	2	10
2-4	75	80	200	30	5	2	15
5-7	200	100	250	40	8	4	20
8-10	320	150	350	60	10	5	20
11-15	450	175	450	80	10	8	25
>15	500	200	500	90	12	10	25

Method of application:

- Before 15-20 days of sapling plantation, full amount of recommended organic and chemical fertilizers should be applied and mixed thoroughly with the soil into the pits followed by irrigation.
- After plantation, fertilizers and manure should be applied in four equal splits at 2-3 months of intervals during March-April, May-June, September and October. Fertilizer should be applied around the tree up to canopy spread leaving 50-100 cm from the tree base and mixed thoroughly with the soil followed by irrigation. Under hill slope conditions, drilling method should be followed.
- For the highly acidic soils, dolomite should be applied at the rate of 250g/tree every 3-4 years interval.

CARAMBOLA (*Averrhoa carambola*)

(Var. BARI Kamranga-1, BARI Kamranga-2, BAU Kamranga-1,
BAU Kamranga-2, BAU Kamranga-3)

Yield Goal: 250± 25 kg/tree/year (above 10 years old tree)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)						OF (kg/tree/year)
	N	P	K	S	Zn	B	
Before planting (pit)	-	60	-	-	-	-	15
0-1	100	50	100	15	0	2	10
2-4	200	75	150	20	5	5	15
5-7	250	100	200	35	5	5	20
8-10	350	120	250	40	8	8	20
>10	450	130	275	50	10	8	25

Method of application:

- Before 15-20 days of sapling plantation, full amount of manure and P should be applied and mixed thoroughly with the soil into the pit, followed by irrigation.
- After plantation, fertilizers and manure should be applied in two equal splits. First installment should be applied during April-May, and the second installment should be applied during September-October. Fertilizer should be applied around the tree up to canopy spread leaving 1.00-1.50 m from the tree base and mixed thoroughly with the soil followed by irrigation. Under hill slope conditions, drilling method should be followed.

JAMUN (*Syzygium cumini*)

(BARI Jam-1)

Yield Goal: 170 ± 17 kg/tree (above 20 years old tree)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)						OF (kg/tree/year)
	N	P	K	S	Zn	B	
Before planting (pit)	-	30	-	-	-	-	10
0-1	75	30	50	15	5	2	10
2-4	150	50	120	30	5	2	15
5-7	250	75	200	40	8	4	20
8-10	350	120	275	50	10	5	20
11-15	400	150	350	60	10	8	25
16-20	450	180	400	70	12	10	25
>20	500	200	450	80	12	10	25

Method of application:

- Before 15-20 days of sapling plantation, total amount of manure and P should be applied and mixed thoroughly with the soil in the pit, followed by irrigation.
- After plantation fertilizers and manure should be applied in two equal splits. First installment should be applied during April-May, and the second installment should be applied during September-October. Fertilizer should be applied around the tree up to canopy spread leaving 1.00-1.50 m from the tree base and mixed thoroughly with the soil followed by irrigation. Under hill slope conditions, drilling method should be followed.

BURMESE GRAPE (*Baccaurea ramiflora*)

(Var: BARI Lotkon-1, BAU Lotkon-1)

Yield Goal: 150 ± 15 kg/tree (above 15 years old tree)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)						OF (kg/tree/year)
	N	P	K	S	Zn	B	
Before planting (pit)	-	50	50	-	-	-	15
0-1	50	20	50	20	5	2	10
2-4	150	60	150	35	5	2	15
5-7	225	80	180	45	8	4	20
8-10	350	100	250	65	10	6	25
11-15	400	120	300	75	12	8	30
>15	450	140	350	100	12	10	35

Method of application:

- Before 15-20 days of sapling plantation, whole amount of manure, P and K should be applied and mixed thoroughly with the soil into the pits, followed by irrigation.
- After plantation fertilizers and manure should be applied in three equal splits. First installment should be applied after fruit harvest. The second and third installments should be applied during October and March, respectively. Fertilizer should be applied around the tree up to canopy spread leaving 50-100 cm from the tree base and mixed thoroughly with the soil followed by irrigation. Under hill slope conditions, drilling method should be followed.

AONLA (*Emblica officinalis*)

(Var. BARI Amloki-1, BARI Amloki-2, BAU Amloki-1)

Yield Goal: 130± 13 kg/tree/year (above 15 years old tree)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)						OF (kg/tree/year)
	N	P	K	S	Zn	B	
Before planting (pit)	-	30	-	-	-	-	10
0-1	100	30	75	18	5	2	15
2-4	220	50	150	35	5	2	15
5-7	300	100	250	45	8	4	20
8-10	400	150	350	50	10	5	25
11-15	500	200	400	55	11	8	25
>15	675	250	500	60	12	10	30

Method of application:

- Before 15-20 days of sapling plantation, full amount of manure and P should be applied and mixed thoroughly with the soil into the pits, followed by irrigation.
- After plantation fertilizers and manure should be applied in two equal splits. First installment should be applied during June-July and second installment should be applied during September-October. Fertilizer should be applied around the tree up to canopy spread leaving 50-100 cm from the tree base and mixed thoroughly with the soil followed by irrigation. Under hill slope conditions, drilling method should be followed.

WAX APPLE (*Syzygium samrangense*)

(Var. BARI Jamrul-1, BAU Jamrul-1, BARI Jamrul-2, BAU Jamrul-2 and BAU Jamrul-3)

Yield Goal: 120 ± 12 kg/tree (above 10 years old tree)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)						OF (kg/tree/year)
	N	P	K	S	Zn	B	
Before planting (pit)	-	35	-	-	-	-	10
0-1	92	40	75	18	4.2	2.6	15
2-4	175	100	150	27	4.2	2.6	15
5-7	230	130	175	36	8.4	4.3	20
8-10	276	150	225	45	10.5	5.1	25
>10	336	200	250	59	10.5	6.8	30

Method of application:

- Before 15-20 days of sapling plantation, full amount of manure and P should be applied separately and mixed thoroughly with the soil in the pit, followed by irrigation.
- After plantation fertilizers and manure should be applied in two equal splits. First installment should be applied during May-June and the second installment should be applied during September-October. Fertilizer should be applied around the tree up to canopy spread leaving 50-100 cm from the tree base and mixed thoroughly with the soil followed by irrigation. Under hill slope conditions, drilling method should be followed.

BAEL (*Aegle marmelos*)

(BARI Bel-1, BARI Bel-2)

Yield Goal: 500 ± 50 fruits/tree (above 20 years old tree)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)						OF (kg/tree/year)
	N	P	K	S	Zn	B	
Before planting (pit)	-	30	-	-	-	-	10
0-1	100	30	75	20	5	2	15
2-4	200	100	150	30	5	2	15
5-7	250	120	200	40	8	4	20
8-10	300	150	250	50	10	5	25
>10	350	200	300	60	10	8	30

Method of application:

- a) Before 15-20 days of sapling plantation, entire amount of manure and P should be applied separately and mixed thoroughly with the soil in the pit, followed by irrigation.
- b) After plantation fertilizers and manure should be applied in two equal splits. First installment should be applied during May-June and the second installment should be applied during September-October. Fertilizer should be applied around the tree up to canopy spread leaving 50-100 cm from the tree base and mixed thoroughly with the soil followed by irrigation. Under hill slope conditions, drilling method should be followed.

WOOD APPLE (*Feronia limonia*)

(Var. BARI Kodbel-1, BARI Kodbel-2, BAU Kodbel-1)

Yield Goal: 300 ± 30 fruits/tree (above 20 years old tree)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)						OF (kg/tree/year)
	N	P	K	S	Zn	B	
Before planting (pit)	-	30	-	-	-	-	10
0-1	100	40	75	15	5	2	10
2-4	200	50	150	35	5	2	15
5-7	250	70	200	50	8	4	20
8-10	300	80	250	70	10	5	20
11-15	350	100	275	75	10	8	25
>15	400	150	300	80	12	10	30

Method of application:

- a) Before 15-20 days of sapling plantation, whole amount recommended organic and chemical fertilizers should be applied and mixed thoroughly with the soil into the pits followed by irrigation.
- b) After plantation fertilizers and manure should be applied in three equal splits during February, May and October. Fertilizer should be applied around the tree up to canopy spread leaving 50-100 cm from the tree base and mixed thoroughly with the soil followed by irrigation. Under hill slope conditions, drilling method should be followed.

BULLOCK'S HEART (*Anona squamosa*) (Ata phal)

(BARI Ata-1)

Yield Goal: 60 ± 6 kg/tree (above 15 years old tree)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)						OF (kg/tree/year)
	N	P	K	S	Zn	B	
Before planting (pit)	-	30	-	-	-	-	10
0-1	100	40	75	15	5	2	10
2-4	200	50	150	35	5	2	15
5-7	250	70	200	50	8	4	20
8-10	300	80	250	70	10	5	20
11-15	350	100	275	75	10	8	25
>15	400	150	300	80	12	10	30

Method of application:

- a) Before 15-20 days of sapling plantation, entire amount of recommended organic and chemical fertilizers should be applied and mixed thoroughly with the soil into the pits followed by irrigation.
- b) After plantation fertilizers and manure should be applied in three equal splits during February, May and October. Fertilizer should be applied around the tree up to canopy spread leaving 50-100 cm from the tree base and mixed thoroughly with the soil followed by irrigation. Under hill slope conditions, drilling method should be followed.

CUSTARD APPLE (*Annona squamosa*) Sharifa)

(All cultivars)

Yield Goal: 50 ± 5 kg/tree (above 10 years old tree)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)						OF (kg/tree/year)
	N	P	K	S	Zn	B	
Before planting (pit)	-	30	-	-	-	-	10
0-1	75	30	75	10	5	2	10
2-4	150	50	120	20	5	2	15
5-7	200	60	150	30	8	4	20
8-10	250	70	175	40	10	5	20
>10	300	80	200	50	10	8	25

Method of application:

- a) Before 15-20 days of sapling plantation, full amount recommended organic and chemical fertilizers should be applied and mixed thoroughly with the soil in the pit followed by irrigation.
- b) After plantation fertilizers and manure should be applied in three equal splits during February, May and October. Fertilizer should be applied around the tree up to canopy spread leaving 50-100 cm from the tree base and mixed thoroughly with the soil followed by irrigation. In hill slope conditions, drilling method should be followed.

VELVET APPLE (*Diospyros discolor*)

(Var. BARI Bilatigab-1, BAU Bilatigab-1, BAU Seedless Bilatigab-2)

Yield Goal: 150 ± 15 kg/tree/year (above 15 years old tree)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)						OF (kg/tree/year)
	N	P	K	S	Zn	B	
Before planting (pit)	-	30	-	-	-	-	10
0-1	75	40	75	10	5	2	10
2-4	150	60	150	20	5	2	15
5-7	250	80	250	40	8	4	20
8-10	350	90	350	50	10	5	20
11-15	450	100	450	55	10	8	25
>15	550	110	550	60	12	10	25

Method of application:

- Before 15-20 days of sapling plantation, whole amount of manure and P should be applied separately and mixed thoroughly with the soil into the pits, followed by irrigation.
- After plantation, fertilizers and manure should be applied in three equal splits during February, May and October. Fertilizer should be applied around the tree up to canopy spread leaving 50-100 cm from the tree base and mixed thoroughly with the soil followed by irrigation. In hill slope conditions, drilling method should be followed.

TAMARIND (*Tamarindus indicus*)

(Var. BARI Tentul-1)

Yield Goal: 100 ± 10 kg/tree/Year (above 20 years old tree)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)						OF (kg/tree/year)
	N	P	K	S	Zn	B	
Before planting (pit)	-	30	-	-	-	-	10
0-1	75	50	75	15	5	2	10
2-4	150	80	200	20	5	2	15
5-7	250	100	300	30	8	4	20
8-10	350	150	400	40	10	5	20
11-15	450	200	500	50	10	8	25
16-20	500	225	550	55	12	10	25
>20	550	250	600	60	12	10	30

Method of application:

- Before 15-20 days of sapling plantation, total amount of manure and P should be applied separately and mixed thoroughly with the soil into the pits, followed by irrigation.
- After plantation fertilizers and manure should be applied in three equal splits. First installment should be applied during May-June, and the second and third installments should be applied during September-October and February-March, respectively. Fertilizer should be applied around the tree up to canopy spread leaving 1.00-1.50 m from the tree base and mixed thoroughly with the soil followed by irrigation. In hill slope conditions, drilling method should be followed.

TAMARIND (*Tamarindus indicus*)

(Var. BAU Tatul-1 (Misti), BAU Tatul-2 (Sour))

Yield Goal: 30 ± 3 kg/tree/Year (above 8 years old tree)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)				OF (kg/tree/year)
	N	P	K	S	
Before planting (in Pit)	-	-	-	-	7
1-2	60	30	75	15	-
3-4	80	40	100	20	7
5-6	100	50	125	25	9
7-8	140	60	150	30	10
>8	200	80	200	40	12

Method of application: Same as above

INDIAN DILLENIA (*Dillenia indica*) (Chalta)

(All cultivars)

Yield Goal: 150 ± 15 kg/tree (above 20 years old tree)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)						OF (kg/tree/year)
	N	P	K	S	Zn	B	
Before planting (pit)	-	30	-	-	-	-	10
0-1	75	30	50	15	5	2	10
2-4	150	50	100	35	5	2	15
5-7	200	75	150	50	8	4	20
8-10	250	90	200	50	10	5	20
11-15	300	100	225	60	10	8	25
>15	350	110	250	70	12	10	25

Method of application:

- a) Before 15-20 days of sapling plantation, full amount of manure and P should be applied separately and mixed thoroughly with the soil into the pits, followed by irrigation.
- b) After plantation fertilizers and manure should be applied in two equal splits. First installment should be applied during April-May, the second installment should be applied during September-October. Fertilizer should be applied around the tree up to canopy spread leaving 1.00-1.50 m from the tree base and mixed thoroughly with the soil followed by irrigation. In hill slope conditions, drilling method should be followed.

DRAGON FRUIT (*Hylocereus undatus*)

(Var. BARI Dragonphol-1, BAU Dragon-1, BAU Dragon-2, BAU Dragon-3, BAU Dragon-4)

Yield Goal: 70 ± 7 kg/plant/year (above 10 years)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)						OF (kg/tree/year)
	N	P	K	S	Zn	B	
Before planting (pit)	0	30	20	15	0	0	20
0-1	75	50	150	15	0	0	10
2-4	150	75	200	25	5	2	15
5-7	175	100	250	30	5	5	20
8-10	200	125	300	40	8	5	20
>10	250	150	350	50	10	8	25

Method of application:

- Before 15-20 days of sapling plantation, total amount of recommended organic and chemical fertilizers should be applied and mixed thoroughly with the soil into the pits followed by irrigation.
- After plantation fertilizers and manure should be applied in two equal splits. First installment should be applied during April-May, the second installment should be applied during September-October. Fertilizer should be applied around the plant base and mixed thoroughly with the soil followed by irrigation.

STRAWBERRY (*Fragaria ananassa*)

(Var. BARI Strawberry-1, BARI Strawberry-2, BARI Strawberry-3, BAU Strawberry-1)

Yield Goal: 10± 1.0 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)						OF (t/ha)
	N	P	K	S	Zn	B	
Optimum	0-45	0-18	0-42	0-9	—	—	5
Medium	46-90	19-36	43-84	10-18	0.0-1.5	0.0-0.7	
Low	91-135	37-54	85-126	19-27	1.6-3.0	0.8-1.4	
Very low	136-180	55-72	127-168	28-36	3.1-4.5	1.5-2.1	

Method of application:

- a) All of OF, P, S, Zn and B; and half of K should be applied as basal during final land preparation.
- b) Nitrogen and half of K should be applied in 4-5 installments at 15-20 day intervals starting from 15 DAP.

LONGAN (*Nephelium longana*)

(Var. BARI Longan-1, BARI Longan-2, BAU Longan-1, BAU Longan-2)

Yield Goal: 50 ± 5 kg/tree/year (above 10 years old tree)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)						OF (kg/tree/year)
	N	P	K	S	Zn	B	
Before planting (pit)	0	30	20	0	0	0	10
0-1	150	50	100	15	0	2	10
2-4	200	75	150	20	5	5	15
5-7	250	100	200	35	5	5	20
8-10	350	150	300	40	8	8	20
>10	400	200	350	50	10	8	25

Method of application:

- a) Before 15-20 days of sapling plantation, total amount of recommended organic and chemical fertilizers should be applied and mixed thoroughly with the soil into the pits followed by irrigation.
- b) After plantation fertilizers and manure should be applied in two equal splits. First installment should be applied during April-May, the second installment should be applied during September-October. Fertilizer should be applied around the tree up to canopy spread leaving 50-100 cm from the tree base and mixed thoroughly with the soil followed by irrigation. Under hill slope conditions, drilling method should be followed.

CASHEWNUT (*Anacardium occidentale*)

(Var. BAU Kajubadam-1)

Yield Goal: 20 ± 2 kg/tree/Year (above 10 years old tree)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)						OF (kg/tree/year)
	N	P	K	S	Zn	B	
Before planting (pit)	-	-	-	-	-	-	10
0-1	75	30	70	-	-	2	10
2-4	90	40	85	15	5	5	15
5-7	105	50	100	20	5	5	20
8-10	125	60	120	25	8	8	20
>10	145	70	140	30	10	8	25

Method of application:

- Before 15-20 days of sapling plantation, all amounts of manure should be applied and mixed thoroughly with the soil in the pit, followed by irrigation.
- After plantation fertilizers and manure should be applied in two equal splits. First installment should be applied during April-May, second installment should be applied during September-October. Fertilizer should be applied around the tree up to canopy spread leaving 50-100 cm from the tree base and mixed thoroughly with the soil followed by irrigation. Under hill slope conditions, drilling method should be followed.

STAR GOOSE BERRY (*Phyllanthus acidus*)

(Var. BAU Orbiroi-1)

Yield Goal: 30 ± 3 kg/plant/Year (above 8 years old tree)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)						OF (kg/plant/year)
	N	P	K	S	Zn	B	
Before planting (in pit)	-	-	-	-	-	-	7
1-2	50	30	100	10	-	1.0	-
3-4	75	50	120	15	5	1.5	7
5-6	100	50	150	20	5	2.0	10
7-8	125	60	200	25	8	2.5	12
>8	150	70	250	30	10	2.5	15

Method of application:

- a) Before planting, recommended OF should be applied in pit as basal dose 15-20 days before planting of sapling and mixed thoroughly with the soil followed by irrigation.
- b) For growing trees, the recommended fertilizers should be applied in two equal splits during June-July and September-October. Fertilizer should be applied around the tree up to the canopy spread leaving 1.0-1.5 m from the tree base and mixed thoroughly with the soil followed by irrigation.

Fig (*Ficus carica*)**(Var. BAU Dumur-1)****Yield Goal: 5 ± 0.5 kg/plant/Year (above 8 years old tree)**

Age of tree (Year)	Nutrient Recommendation (g/tree/year)					OF (kg/plant/year)
	N	P	K	S	B	
Before planting (in pit)	-	-	-	-	-	7
1-2	50	30	70	10	1.0	-
3-4	90	40	85	15	1.5	7
5-6	105	50	100	20	2.0	10
7-8	125	60	120	25	2.5	12
>8	145	70	140	30	2.5	15

Method of application:

- a) Before planting, recommended OF should be applied into the pits as basal dose 15-20 days before planting of sapling and mixed thoroughly with the soil followed by irrigation.
- b) For growing trees, the recommended fertilizers should be applied in two equal splits during June-July and September-October. Fertilizer should be applied around the tree up to the canopy spread leaving 1.0-1.5 m from the tree base and mixed thoroughly with the soil followed by irrigation.

INDIAN OLIVE (*Elaeocarpus floribundus*)

(Var. BARI Jalpai-1, BAU Jalpai-1)

Yield Goal: 150 ± 15 kg/tree/Year (above 20 years old tree)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)						OF (kg/plant/year)
	N	P	K	S	Zn	B	
Before planting (pit)	-	30	-	-	-	-	10
0-1	75	30	50	10	5	2	10
2-4	100	50	90	20	5	2	15
5-7	150	75	120	30	8	3	20
8-10	200	80	150	40	10	3	20
11-15	250	85	200	50	10	4	25
>15	300	90	220	60	12	4	25

Method of application:

- Before 15-20 days of sapling plantation, total amount of manure and P should be applied separately and mixed thoroughly with the soil into the pit, followed by irrigation.
- After plantation fertilizers and manure should be applied in two equal splits. The first installment should be applied during April-May, the second installment should be applied during September-October. Fertilizer should be applied around the tree up to canopy spread leaving 1.00-1.50 m from the tree base and mixed thoroughly with the soil followed by irrigation. Under hill slope conditions, drilling method should be followed.

WATERMELON (*Citrullus colocynthis*)

(Var. BARI Tormuj-1, BARI Tormuj-2)

Yield Goal: 50.0 ± 5.0 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)							OF (t/ha)
	N	P	K	S	Mg	Zn	B	
Optimum	0-39	0-12	0-40	0-5	0-3	-	-	7
Medium	40-78	13-24	41-80	6-10	4-6	0-1.6	0-1.0	
Low	79-117	25-36	81-120	11-15	7-9	2.3-3.2	1.1-2.0	
Very low	118-156	37-48	121-160	16-20	10-12	4.5-4.8	2.1-3.0	

Method of application:

- One-third of the OF should be applied during final land preparation.
- Rest two-thirds of the OF and all of P, S, Mg, Zn, B and one-third of K should be applied as basal into the pits.

- c) One-fifth of N and one-third of K should be applied after 20 and 40 days of transplanting. After that, the rest of N could be applied in equal two installments at 20-day intervals.

NETTED MELON

(Variety: OP, Hybrid)

Yield Goal: 20.0 ±2.0 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)						OF (t/ha)
	N	P	K	S	Zn	B	
Optimum	0-20	0-11	0-30	0-5	-	-	5
Medium	21-40	12-22	31-60	6-10	0-1.6	0.0-1.0	
Low	41-60	23-33	61-90	11-15	2.3-3.2	1.1-2.0	
Very low	61-80	34-44	91-120	16-20	4.5-4.8	2.1-3.0	

Method of application: Same as watermelon.

7.9 Hydroponic Culture

Fertilizer Recommendation (Amount: g/1000 L)*										
Ca(NO ₃) ₂ . 4 H ₂ O	HNO ₃	KOH	MgSO ₄ . 7 H ₂ O	KH ₂ PO ₄	H ₃ BO ₃	ZnSO ₄ . 7 H ₂ O	MnSO ₄ . 5 H ₂ O	CuSO ₄ . 5 H ₂ O	(NH ₄) ₆ Mo ₇ O ₂₄ . 4 H ₂ O	Fe-EDTA
1000	272 ml	275	510	270	1.80	0.44	6.10	0.40	0.38	80

*BARI Hydroponic Solution-1; Electrical conductivity (EC) of Full strength of BARI Hydroponic Solution = 2.3~2.5 dS/m. The solution can be used in the following crops:

Crop	Variety	Yield	Application method
Tomato (<i>Solanum lycopersicum</i>)	BARI Tomato – 14	28.0 ± 2.0 kg/m ²	At seedling stage, low concentration of BARI Hydroponic Solution – 1 (1.5 dS/m EC) should be applied until transplanting. After transplanting till anthesis, hydroponic nutrient solution concentration should be maintained at 1.5 dS/m, while after anthesis and fruit development stage it should be increased to 2.3-2.5 dS/m and during fruit ripening stage EC should be maintained around 2.0 dS/m.
Sweet pepper (<i>Capsicum annuum</i> L.)	BARI Mistimorich – 1, and – 2	50.0 ± 5.0 t/ha	At seedling stage, low concentration of BARI Hydroponic Solution – 1 (1.5 dS/m EC) should be applied until transplanting. After transplanting till anthesis, hydroponic nutrient solution concentration should be maintained at 1.5 dS/m, while after anthesis and fruit development stage till harvest it should be increased to 2.0 dS/m
Lettuce (<i>Lactuca sativa</i>)	BARI Lettuce – 1	15.0 ± 1.0 kg/m ²	At seedling stage, low concentration of BARI Hydroponic Solution – 1 (1.5 dS/m EC) should be applied until transplanting. After transplanting till

Crop	Variety	Yield	Application method
			anthesis, hydroponic nutrient solution concentration should be maintained at 1.5-2.0 dS/m throughout the growing period until harvest.
Strawberry (<i>Fragaria ananassa</i> Duch.) ×	BARI Strawberry – 1, – 2, and – 3	15.0 ± 1.0 kg/m ²	After transplanting till harvest, hydroponic nutrient solution concentration should be maintained at 1.0-1.2 dS/m.

7.10 Rooftop Gardening

MANGO (*Mangifera indica*)

(Var. BARI Aam-3, BARI Aam-4, BARI Aam-11)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)						OF (kg/plant/year)
	N	P	K	S	Zn	B	
Before planting (pot)	0	30	0	0	0		0
1-2	56	25	35	10	4	3	10
3-4	70	32	40	15	6	5	10
5-6	85	40	45	20	8	5	10
7-8	95	45	50	25	10	8	10
>9	120	60	75	35	12	10	10

Method of application:

- Before 15-20 days of sapling plantation, full amount of manure and P should be applied and mixed thoroughly with the soil into the pot, followed by irrigation.
- Pot size: 18-20" x 18 -20"
- Media: After plantation, fertilizers should be applied in four equal splits. The first, second, third and fourth installments should be applied during September, December, March and June, respectively. Manure should be applied in two splits during September and March. However, refilling of the media should be done if necessary.

DRAGON FRUIT

(Var. BARI Dragonphol-1, BAU Dragon-1, BAU Dragon-2, BAU Dragon-3, BAU Dragon-4)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)						OF (kg/plant/year)
	N	P	K	S	Zn	B	
0–1	46	14	36	7	3.5	0.87	25
2–3	55	18	48	9.5	5	1.40	06
4–6	64	20	54	12	7	1.75	09
>6	69	24	60	12	7	1.75	12

Method of application:

- Before planting (1st year) 20 kg organic manure and 1/6th of other inorganic fertilizers should be applied to the potting media (70-100 kg soil/half drum).
- Afterward, the rest of the fertilizers will be applied in 5 splits at two months interval. From 2nd year, all the fertilizers are suggested to be applied in six splits at two months interval.
- Fertilizers should be applied in the periphery of the pot followed by light irrigation.

LEMON

(Var. BARI Lebu-1, BARI Lebu-2, BARI Lebu-3, BAU Lemon-1,
BAU Lemon-2, BAU Lemon-3, BAU Lemon-4)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)						OF (kg/plant/year)
	N	P	K	S	Zn	B	
0-1	18	6	18	2.5	1.0	0.87	25
2-3	23	8	24	3.0	2.0	0.87	06
4-6	28	9	24	3.5	3.0	1.39	09
>6	28	9	24	3.5	4.0	1.75	12

Method of application:

- Before planting (1st year) 20 kg organic manure and 1/6th of other inorganic fertilizers should be applied to the potting media (70-100 kg soil/half drum).
- Afterward, the rest of the fertilizers need to be applied in 5 splits in 2 month intervals. From the 2nd year, all the fertilizers are suggested to be applied in six splits at 2 month intervals.
- Fertilizers should be applied in the periphery of the pot followed by light irrigation.

GUAVA

(Var. BARI Payara-1 (Kazi Payara), BARI Payara-2, BARI Payara-3,
BAU Payara-7, BAU Payara-8, BAU Payara-9)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)						OF (kg/plant/year)
	N	P	K	S	Zn	B	
0-1	19	6	18	2.35	1.65	7.00	25
2-3	23	7	21	2.80	2.64	9.00	06
4-6	28	8	24	3.50	3.30	10.50	09
>6	30	9	27	3.50	4.00	11.50	12

Method of application:

- Before planting (1st year) 20 kg organic manure and 1/6th of other inorganic fertilizers should be applied to the potting media (70-100 kg soil/half drum).
- Afterward, the rest of the fertilizers need to be applied in 5 splits in 2- month intervals. From the 2nd year, all the fertilizers are suggested to be applied in six splits at 2-month intervals.
- Fertilizers should be applied in the periphery of the pot followed by light irrigation.

GOLDEN APPLE

(BARI Amra-1, BAU Amra-1)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)						OF (kg/plant/year)
	N	P	K	S	Zn	B	
0-1	14	8	18	3.53	1.65	0.87	25
2-3	19	10	24	4.70	2.64	1.39	06
4-6	23	12	30	4.70	2.64	1.39	09
>6	28	14	36	6.00	3.30	1.39	09

Method of application:

- Before planting (1st year) 20 kg organic manure and 1/6th of other inorganic fertilizers should be applied to the potting media (70-100 kg soil/half drum).
- Afterwards, the rest of the fertilizers need to be applied in 5 splits in 2- month intervals. From the 2nd year, all the fertilizers are suggested to be applied in six splits at 2- month intervals.
- Fertilizers should be applied in the periphery of the pot followed by light irrigation.

7.11 Floating Agriculture**Fertilizer management**

The floating beds should be prepared as described in the text part of this Guide. The crops grown in the floating beds should be fertilized at the rates given in the table below. All of the fertilizers should be applied in liquid forms in six equal instalments for bottle gourd, turmeric and chilli and in five equal instalments for the other crops as mentioned in the table. The fertilizers should be applied at 15 days' intervals in turmeric and at 10 days' intervals in other crops since 15 days after seedling transplanting. Fertilizers should be dissolved in 10 litres of water and be applied in the bed around the plants. Zinc fertilizer should be applied as foliar spray. After heavy rainfall the rate of fertilizers should slightly be increased. Urea should not be applied as over dose which enhances decomposition of the bed and reduces its durability. There is no need of fertilizer application for raising seedlings on the floating bed because it may make the seedlings weaker and increase their mortality.

Fertilizer recommendation for vegetables and spices in floating bed

Vegetable/spice crops	Recommendation (g/10 sqm floating bed)					
	N	P	K	S	Zn	B
A. Vegetable crops						
1. Cucumber	26	13	9	5.0	0.42	0.77
2. Bottle gourd	24	27	15	5.9	0.42	0.77
3. Pumpkin	24	27	15	5.9	0.42	0.77
4. Watermelon	32	19	14	4.3	0.42	0.77
5. Bitter gourd	21	16	16	5.4	-	0.77
6. Yard long bean	20	16	16	5.0	0.42	0.77
7. Okra	42	16	12	4.3	0.42	0.77
8. Summer tomato	56	34	19	4.3	0.42	0.77
B. Spice crop						
9. Turmeric	44	18	53	8.5	-	1.12
10. Chilli	22	43	24	4.3	0.42	0.77

7.12 Plantation Crops

SUGARCANE (*Saccharum officinarum*)

(Var. BSRI Akh 41, BSRI Akh 42, BSRI Akh 43, BSRI Akh 44, BSRI Akh 45, BSRI Akh 46, BSRI Akh 47, BSRI Akh 48)

Yield Goal: 100 ± 10 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)						OF* (t/ha)
	N	P	K	S	Zn	B	
Optimum	0-54	0-18	0-53	0-6	-	-	3
Medium	55-108	19-36	54-106	7-12	0-1.6	0-1.0	
Low	109-162	37-54	107-159	13-18	1.7-3.2	1.1-2.0	
Very low	163-220	55-72	160-212	19-24	3.3-4.8	2.1-3.0	

Method of application:

- For heavy textured soils, half of N and K and all of P, S, Mg and Zn should be applied in trench and mixed thoroughly with the soil before planting of sugarcane. Remaining N and K should be applied as top dress at tillering stage (120–150 days after planting).
- For light textured soils, one-third of N and K, and all of P, S, Mg and Zn should be applied in trench and mixed thoroughly with the soil before planting of sugarcane. Remaining N and K should be applied in two equal splits as top dress at tillering stage (90–120 DAP) and after completion of tillering (150-180 DAP).
- Top dressing of N and K in both cases (a and b) should be done under moist soil condition. If the soil is dry, top dressing would be delayed for rainfall.
- For transplanted sugarcane basal N should be applied at 20–30 DAT.
- For ratoon cane additional 40 kg N/ha should be applied and all other nutrients should be same as of the plant cane.

SUGAR BEET (*Beta vulgaris*)

(All varieties)

Yield Goal: 80 ± 8 t/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)						OF
	N	P	K	S	Mg	Zn	
Optimum	0–32	0–16	0–40	0–6	0–3	–	5
Medium	33–64	17–32	41–80	7–12	4–6	0–2	
Low	65–96	33–48	81–120	13–18	7–9	3–4	
Very low	97–148	49–56	121–160	19–24	10–12	5–6	

Methods of application:

- All of OF, P, S, Mg and Zn; and one-third of N and K should be applied as basal during final land preparation.
- Remaining N and K should be side dressed in two equal splits at 30 and 60 days after sowing under moist soil condition and mixed thoroughly with the soil as soon as possible for better utilization.

PALMYRA PALM (*Borassus flabellifer*) (Tal)

(All cultivars)

Yield Goal: 200 ± 20 fruits/tree (above 20 years old tree)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)			OF (kg/tree/year)
	N	P	K	
Before Planting (in Pit)	-	-	-	7
1–4	100	40	100	5
5–10	200	50	200	7
11–15	300	60	300	10
16–20	400	80	400	12
> 20	500	100	500	15

Method of application:

- Before planting, recommended OF should be applied into the pits as basal at 15–20 days before planting of sapling and mixed thoroughly with the soil followed by irrigation.
- For growing trees, the recommended fertilizers should be applied in two equal splits during April–May and September–October. Fertilizer should be applied around the tree and mixed thoroughly with the soil followed by irrigation.

TEA (*Camellia sinensis*)

(All varieties)

Young Tea

Age of Plant (year)	Nutrient Recommendation (kg/ha)			OF (kg/plant)
	N	P	K	
1	80	40	80	2
2	90	45	90	2
3	120	40	80	3
4	135	45	90	5
5	150	50	100	5

Method of application:

Fertilizers should be applied in three split doses

- 1st split : April – May
- 2nd split : August
- 3rd split : October – November

Mature tea*

Yield (kg/ha)	Nutrient Recommendation (kg/ha)						
	1 st Split				2 nd Split in soil (NK) & Foliar spray (Zn)		
	N	P	K	Zn	N	K	Zn (g/200 lit water per ha)
Up to 1000	50	15	25	—	—	—	—
1001–2000	51–100	15-20	26-50	2	60	20	840
2001–3000	101–150	20-25	51-75	3	60	20	840

*Fertilizer recommendation has been estimated on the basis of production of made tea in kg/ha

Method of application:

- All of P and Zn; and half of N and K should be applied during April–May with onset of shower. Remaining half of N and K should be applied within July to first week of August.
- Urea is generally used in tea cultivation as a source of nitrogen; and TSP/Rock phosphate/DAP and MOP might be used as the sources of P and K, respectively.
- Zinc should also be applied as foliar spray four times a year- twice during mid-April to Mid-June and twice during mid-September to mid-October.
- If soil pH is below 5.0, dolomite [(CaMg(CO₃)₂)] application at 1-2 t/ha is suggested, usually after a good shower.

Nucleus Clone Plot

The tea plants which are nourished for vegetative cuttings.

Fertilizer Recommendation for Nucleus Clone Plot

Age of plant (year)	NPK ratio	Mixture (g/bush)	Fertilizer mixture (kg/ha)		Method of application
			Planting geometry		
			Square (6944 bushes/ha)	Triangular (8019 bushes/ha)	
1	2:1:2	30	208	240	Ring
2	2:1:2	50	347	400	Ring
3	2:1:2	70	485	560	Ring
4	2:1:2	80	530	630	Ring
> 4	2:1:2	80	530	630	Broadcast

Method of application:

- Fertilizers should be applied in two equal splits annually.
- Rate of N could be 90 kg/ha for the first three years; and from the fourth year onwards N rate could be 100 kg/ha

Seed Bari

A seed garden is popularly known as seed bari and the plants are not plucked. The plants are reared for collection of tea seeds.

Fertilizer Recommendation for Seed Bari

Age of Plant (Year)	Urea (g/plant)	TSP (g/plant)	MOP (g/plant)	Cowdung (kg/plant)	Oilcake (kg/plant)
1	50	50	50	5	0.5
2	100	100	100	5	0.5
3	150	300	300	10	1.0
4	200	400	400	10	1.0
> 4	250	500	500	10	1.0

Method of application:

- Fertilizers should be applied in two equal splits annually.
- Rate of N could be 90 kg/ha for the first three years; and from the fourth year onwards N rate could be 100 kg/ha

BETEL LEAF (*Piper betle*)

(Var. BARI Pan-1, BARI Pan-2, BARI Pan-3)

Yield Goal: 35± 3.5 lac leaf/ha/year

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha/year)					Oilcake (t/ha/year)
	N	P	K	S	Zn	
Optimum	0–20	0–8	0–12	0–4	-	3
Medium	21–40	9–16	13–24	5–8	0-1.0	
Low	41–60	17–24	25–36	9–12	1.1-2.0	
Very low	61–80	25–32	37–48	13–16	2.1-3.0	

Method of application:

- One-third of oilcake and all of P, K and S should be applied as basal during final land preparation.
- After 30–40 days of planting, remaining oilcake should be applied at 15–20 days of intervals in rows until the vine grows up to 1.0–1.5 meter in length.
- Nitrogen should be applied in 3 equal installments before the monsoon rain and in September and December.
- In the subsequent years, the above fertilizer doses should be applied in 3 equal installments, before the monsoon rains and in the month of September and December.

BETELNUT (*Areca catechu*)

(All cultivars)

Yield Goal: 3.0± 0.3 kg dry nut/tree (Above 11years old tree)

Age of tree (Year)	Nutrient Recommendation (g/tree/year)				OF (kg/tree/year)
	N	P	K	S	
Before planting (in Pit)	-	30	-	18	6
1–2	120	120	135	-	-
3–5	175	165	180	36	5
6–8	225	210	230	42	7
9–11	275	255	280	48	7
> 11	325	300	325	54	10

Method of application:

- All of OF, P and S recommended for 'before planting' application should be applied into the pits as basal dose 15-20 days before planting of seedling and mixed thoroughly with the soil followed by irrigation.
- For growing trees, the recommended fertilizers should be applied in two equal splits during May just after harvesting of fruit and September-October. Fertilizer should be applied around the tree up to the canopy spread leaving 0.5-1.0 m from the tree base and mixed thoroughly with the soil followed by irrigation.

RUBBER (*Hevea brasiliensis*)

(Clone: TJIR-1, RRIM-600, PBIG, PB-217, PB-235, GG)

Yield Goal: 2.7 ± 0.27 t/ha/year

Age of tree (Year)	Nutrient Recommendation (g/tree/year)		
	N	P	K
Before planting	45	90	50
1–7	45	45	50
> 7	56	68	55

Method of application:

- All of N, P and K, recommended for ‘before planting’ application should be applied into the pits 10-15 days before planting of saplings followed by irrigation.
- For growing trees, N and K should be applied in two equal splits during April-May at the onset of monsoon and after monsoon during September-October. Fertilizers should be applied around the tree up to canopy spread leaving 1.0-1.5 m from the tree base and mixed thoroughly with the soil followed by irrigation.
- Phosphorus should be applied in alternate years as broadcast and mixed thoroughly with the soil followed by irrigation.
- In the hill slope, the fertilizers should be applied in dibbling method (4 to 5 holes per tree to a depth 5-8 cm at the upper part of the slope).

7.13 Flowers

MARIGOLD (*Tagetes erecta* L.)

(Var. BARI Gada-1, French marigold)

Yield Goal : 10.0 ± 1.0 ton flowers/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)					OF (t/ha)
	N	P	K	S	Zn	
Optimum	0–30	0–13	0–30	0-5	-	3
Medium	31–60	14–26	31–60	6-10	0-2.0	
Low	61–90	27–39	61–90	11-15	2.1-4.0	
Very low	91–120	40–52	91–120	16-20	4.1-6.0	

Method of application:

- All of OF and chemical fertilizers except N should be applied as basal during final land preparation.
- Nitrogen should be applied in three equal splits at 25, 45 and 65 DAP.

MARIGOLD (*Tagetes erecta* L.)

(African marigold)

Yield Goal: 15± 1.5 ton flowers/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)					OF (t/ha)
	N	P	K	S	Zn	
Optimum	0–45	0–20	0–40	0–6	-	5
Medium	46–90	21–40	41–80	7–12	0–3.0	
Low	91–135	41–60	81–120	13–14	3.1–6.0	
Very low	136–180	61–80	121–160	15–24	6.1–9.0	

Method of application:

- All of OF and chemical fertilizers except N should be applied as basal during final land preparation.
- Nitrogen should be applied in three equal splits at 25, 45 and 65 DAP.

MARIGOLD (*Tagetes erecta* L.)

(BARI Gadha – 1)

(Hydroponic culture)

Yield Goal: 15.0 ± 2.0 flowers/plant

Fertilizer Recommendation (Amount: g/1000 L)*									
Ca(NO ₃) ₂ . 4 H ₂ O	KNO ₃	MgSO ₄ . 7 H ₂ O	KH ₂ PO ₄	H ₃ BO ₃	ZnSO ₄ . 7 H ₂ O	MnSO ₄ . 5 H ₂ O	CuSO ₄ . 5 H ₂ O	(NH ₄) ₆ Mo ₇ O ₂₄ . 4 H ₂ O	Fe-EDTA
1000	580	510	270	1.80	0.44	6.10	0.40	0.38	80

*BARI Hydroponic Solution-1 (Asaduzzaman et al., 2021); EC of Full strength of BARI Hydroponic Solution = 2.3~2.5 dS/m.

Method of application:

At seedling stage, low concentration of BARI Hydroponic Solution – 1 (1.5 dS/m EC) should be applied until transplanting. After transplanting hydroponic nutrient solution concentration should be maintained at 1.5 dS/m till harvest.

ROSE (*Rosa centiflora*)

(Class: Hybrid tea)

Yield Goal : 750,000 – 800,000 flower sticks/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)					OF (t/ha)
	N	P	K	S	Zn	
Optimum	0–35	0–25	0–35	0–5	-	5
Medium	36–70	26–50	36–70	6–10	0–2.0	
Low	71–105	51–75	71–105	11–15	2.1–4.0	
Very low	106–140	76–100	106–140	16–20	4.1–6.0	

Method of application:

- a) All of organic and chemical fertilizers except N should be applied as basal during final land preparation.
- b) Nitrogen should be applied in three equal splits as topdress at 30, 60 and 90 DAP.

TUBEROSE (*Polianthes tuberosa*)

(Var. Double)

Yield Goal: 115,000–125,000 sticks/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)						OF (t/ha)
	N	P	K	S	Zn	B	
Optimum	0–75	0–23	0–44	0–5	–	-	5
Medium	76–150	24–46	45–88	6–10	0.0–2.0	0–1.0	
Low	151–225	47–69	89–132	15–15	2.1–4.0	1.1–2.0	
Very low	226–300	70–92	133–176	16–20	4.1–6.0	2.1–3.0	

Method of application:

- a) All of organic and chemical fertilizers except N should be applied as basal during final land preparation.
- b) Nitrogen should be applied in three equal splits as topdress at 30, 45 and 60 DAP.

GLADIOLUS (*Gladiolus spp.*)

(Var. BARI Gladiolus-3, BARI Gladiolus-4, BARI Gladiolus-5)

Yield Goal: 175,000–200,000 flower sticks/ha

Soil analysis interpretation	Nutrient Recommendation (kg/ha)						OF (t/ha)
	N	P	K	S	Zn	B	
Optimum	0–60	0–22	0–45	0–8	–	–	5
Medium	61–120	23–44	46–90	9–16	0.0–2.0	0.0–0.7	
Low	121–180	45–66	91–135	17–24	2.1–4.0	0.8–1.4	
Very low	181–240	67–88	136–180	25–32	4.1–6.0	1.5–2.1	

Method of application:

- c) All of organic and chemical fertilizers except N should be applied as basal during final land preparation.
- d) Nitrogen should be applied in three equal splits as topdress at 30, 45 and 60 DAP.

ZINNIA (*Zinnia elegans*)

(All varieties)

Yield Goal : 200,000 – 300,000 flower sticks/ha

Soil analysis interpretation	Nutrient Recommendation (kg/ha)					OF (kg/tree/year)
	N	P	K	S	Zn	
Optimum	0–25	0–8	0–15	0–5	–	5
Medium	26–50	9–16	16–30	6–10	0.0–2.0	
Low	51–75	17–24	11–45	11–15	2.1–4.0	
Very low	76–100	25–32	46–180	16–20	4.1–6.0	

Method of application:

- a) All of organic and chemical fertilizers except N should be applied as basal during final land preparation.
- b) Nitrogen should be applied in three equal installments at 25, 45, 65 DAP.

GERBERA (*Gerbera jamesonii*)

(Var. BARI Gerbera-1, BARI Gerbera-2)

Yield Goal: 850,000 – 900,000 flower sticks/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha/year)						OF (t/ha)	Coco-dust (kg/ha)
	N	P	K	S	Zn	B		
Optimum	0-50	0-25	0-40	0-8	-	-	5	1000
Medium	51-100	26-50	41-80	9-16	0-2.0	0-1.0		
Low	101-150	51-75	81-120	17-24	2.1-4.0	1.1-2.0		
Very low	151-200	76-100	121-160	25-32	4.1-6.0	2.1-3.0		

Method of application:

- c) All of organic and chemical fertilizers except N should be applied as basal during final land preparation.
- d) Nitrogen should be applied in three equal installments at 30, 60 and 90 days after planting.

CHRYSANTHEMUM (*Chrysanthemum coronarium*)

(Var. BARI Chandramallika-1, BARI Chandramallika-2, BARI Chandramallika-3, BARI Chandramallika-4, Large flowered varieties)

Yield Goal : 2500,000–3000,000 flowers/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha/year)						OF (t/ha)
	N	P	K	S	Zn	B	
Optimum	0-50	0-25	0-40	0-5	-	-	3
Medium	51-100	26-50	41-80	6-10	0-2.0	0-1.0	
Low	101-150	51-75	81-120	11-15	2.1-4.0	1.1-2.0	
Very low	151-200	76-100	121-160	16-20	4.1-6.0	2.1-3.0	

Method of application:

- a) All of organic and chemical fertilizers except N should be applied as basal during final land preparation.
- b) Nitrogen should be applied in three equal installments at 25, 45 and 65 DAT.

CHRYSANTHEMUM (*Chrysanthemum coronarium*)

(Var. BARI Chandramallika-1, BARI Chandramallika-2, BARI Chandramallika-3, BARI Chandramallika-4, Small flowered varieties)

Yield Goal : 3500,000–4000,000 flowers/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha/year)						OF (t/ha)	Bonemeal (kg/ha)
	N	P	K	S	Zn	B		
Optimum	0–45	0–22	0–40	0–5	-	-	2	15
Medium	46–90	23–44	41–80	6–10	0-2.0	0-1.0		
Low	91–135	45–66	81–120	11–15	2.1-4.0	1.1-2.0		
Very low	136–180	67–88	121–160	16–20	4.1-6.0	2.1-3.0		

Method of application:

- All of organic and chemical fertilizers except N should be applied as basal during final land preparation.
- Nitrogen should be applied in three equal installments at 25, 45 and 65 DAT.

ORCHID (*Phaius tankervilleae*)

(Var. BARI Orchid-1)

Yield Goal : 18,000–20,000 flowers/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha/year)						OF (t/ha)
	N	P	K	S	Zn	B	
Optimum	0–22	0–10	0–20	0–5	-	-	5
Medium	23–44	11–20	21–40	6–10	0-2.0	0-1.0	
Low	45–66	21–30	41–60	11–15	2.1-4.0	1.1-2.0	
Very low	67–88	31–40	61–80	16–20	4.1-6.0	2.1-3.0	

Method of application:

- All of organic and chemical fertilizers except N should be applied as basal during final land preparation.
- Nitrogen should be applied as topdress in three equal splits at 30, 60 and 90DAT and mixed thoroughly with the soil.

GYPSOPHILA (*Gypsophila paniculata*)

(Var. BARI Gypsophila-1)

Yield Goal: Seed-1200 kg/ha, Cut flower bunch yield- 190000 nos. bunch/ha

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha)						OF (t/ha)
	N	P	K	S	Zn	B	
Optimum	0-29	0-16	0-24	0-8	—	—	5
Medium	30-58	17-32	25-48	9-16	0.0–1.3	0.0–0.7	
Low	59-87	33-48	49-72	17-24	1.4–2.6	0.8–1.4	
Very low	88-116	49-64	73-96	25-32	2.7–3.9	1.5–2.1	

Method of application:

- a) The full dose of cowdung, one-third of N, full of P, half of K, and full of S should be applied during final land preparation.
- b) Remaining two-third of N and half of K should be applied in two equal splits, the first split at 20 days after sowing (DAS) and the second split at 35 DAS.

7.14 Fodder Crops

NAPIER GRASS (*Pennisetum purpureum*)

(Var. IGFRI-3, IGFRI-6, IGFRI-7, IGFRI-10, CO-3)

Yield Goal (Green biomass): 300 ± 30 t/ha/year

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha/year)				
	N	P	K	S	Zn
Optimum	0–45	0–6	0–10	0–3	-
Medium	46–90	7–12	11–20	4–6	0-1.3
Low	91–135	13–18	21–30	7–9	1.4-2.6
Very low	136–180	19–24	31–40	10–12	2.7-3.9

Method of application:

- All of P, K, S and Zn and 25% of N should be applied as basal during final land preparation.
- 25% of N should be top dressed after 25 days of sowing/planting.
- The remaining 50% of N should be top dressed in equal splits after each cut at 30 – 40 days' intervals.

PARA GRASS (*Brachiaria mutica*)

Yield Goal (Green biomass): 120 ± 12 t/ha/year

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha/year)				
	N	P	K	S	Zn
Optimum	0–30	0–5	0–8	0–2	-
Medium	31–60	6–10	9–16	3–4	0-1.0
Low	61–90	11–15	17–24	5–6	1.1-2.0
Very low	91–120	16–20	25–32	7–8	2.1-3.0

Method of application:

- All of P, K, S and Zn and 25% of N should be applied as basal during final land preparation.
- 25% of N should be top dressed after 25 days of sowing/planting.
- The remaining 50% of N should be top dressed in equal splits after each cut at 30 – 40 days' intervals.

GERMAN GRASS (*Echinochloa crusgalli*)

Yield Goal (Green biomass): 120 ± 12 t/ha/year

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha/year)				
	N	P	K	S	Zn
Optimum	0–33	0–4	0–6	0–2	-
Medium	34–66	5–8	7–12	3–4	0–1.0
Low	67–99	9–12	13–18	5–6	1.1–2.0
Very low	100–132	13–16	19–24	7–8	2.1–3.0

Method of application:

- a) All of P, K, S and Zn and 25% of N should be applied as basal during final land preparation.
- b) 25% of N should be top dressed after 25 days of sowing/planting.
- c) The remaining 50% of N should be top dressed in equal splits after each cut at 30 – 40 days' intervals.

DHAL GRASS (*Hymenachne amplexicaulis*)

Yield Goal (Green biomass): 100 ± 10 t/ha/year

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha/year)				
	N	P	K	S	Zn
Optimum	0–25	0–4	0–6	0–1.5	-
Medium	26–50	5–8	7–12	1.6–3.0	0–1.0
Low	51–75	9–12	13–18	3.1–4.5	1.1–2.0
Very low	76–100	13–16	19–24	4.6–6.0	2.1–3.0

Method of application:

- a) All of P, K, S and Zn and 25% of N should be applied as basal during final land preparation.
- b) 25% of N should be top dressed after 25 days of sowing/planting.
- c) The remaining 50% of N should be top dressed in equal splits after each cut at 30 – 40 days' intervals.

GUINEA GRASS (*Panicum maximum*)

Yield Goal (Green biomass): 120 ± 12 t/ha/year

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha/year)				
	N	P	K	S	Zn
Optimum	0–33	0–4	0–6	0–1.5	-
Medium	34–66	5–8	7–12	1.6–3.0	0–1.0
Low	67–99	9–12	13–18	3.1–4.5	1.1–2.0
Very low	100–132	13–16	19–24	4.6–6.0	2.1–3.0

Method of application:

- d) All of P, K, S and Zn and 25% of N should be applied as basal during final land preparation.
- e) 25% of N should be top dressed after 25 days of sowing/planting.
- f) The remaining 50% of N should be top dressed in equal splits after each cut at 30 – 40 days' intervals.

JUMBO (Var. Hybrid sorghum)

Yield Goal: 150 ± 15 t/ha/year

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha/year)				
	N	P	K	S	Zn
Optimum	0–37	0–5	0–7	0–2	-
Medium	38–74	6–10	8–14	3–4	0–1.0
Low	75–111	11–15	15–21	5–6	1.1–2.0
Very low	112–148	16–20	22–28	7–8	2.1–3.0

Method of application:

- a) All of P, K, S and Zn and 25% of N should be applied as basal during final land preparation.
- b) 25% of N should be top dressed after 25 days of sowing/planting.
- c) The remaining 50% of N should be top dressed in equal splits after each cut at 30 – 40 days' intervals.

ROZI (*Brachiaria ruzuziensis*)

Yield Goal: 120 ± 12t/ha/year

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha/year)				
	N	P	K	S	Zn
Optimum	0–30	0–4	0–6	0–1.5	-
Medium	31–60	5–8	7–12	1.6–3.0	0-1.0
Low	61–90	9–12	13–18	3.1–4.5	1.1-2.0
Very low	91–120	13–16	19–24	4.6–6.0	2.1-3.0

Method of application:

- a) All of P, K, S and Zn and 25% of N should be applied as basal during final land preparation.
- b) 25% of N should be top dressed after 25 days of sowing/planting.
- c) The remaining 50% of N should be top dressed in equal splits after each cut at 30 – 40 days' intervals.

MAIZE (*Zea mays* L.)

Yield Goal: 80 ± 8 t/ha/year

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha/year)				
	N	P	K	S	Zn
Optimum	0–30	0–4	0–5	0–1.5	-
Medium	31–60	5–8	6–10	1.6–3.0	0-1.0
Low	61–90	9–12	11–15	3.1–4.5	1.1-2.0
Very low	91–120	13–16	16–20	4.6–6.0	2.1-3.0

Method of application:

- a) All of P, K, S and Zn and 50% of N should be applied as basal during final land preparation.
- b) Remaining 50% of N should be top dressed after 30 days of sowing/planting.

OAT (*Avena sativa*)

Yield Goal: 50 ± 5 t/ha/year

Soil Analysis Interpretation	Nutrient Recommendation (kg/ha/year)				
	N	P	K	S	Zn
Optimum	0–15	0–3	0–4	0–1.5	-
Medium	16–30	4–6	5–8	1.6–3.0	0-1.0
Low	31–45	7–9	9–12	3.1–4.5	1.1-2.0
Very low	46–60	10–12	13–16	4.6–6.0	2.1-3.0

Method of application:

- a) All of P, K, S and Zn and 50% of N should be applied as basal during final land preparation.
- b) Remaining 50% of N should be top dressed after 30 days of sowing/planting.